

Independent external validation of the SugarCode AI computational-biology platform against held-out public benchmarks

Udita Phookan

September 25, 2026

Independent research; computational work assisted by Instinct

Abstract

We subject SugarCode AI, a 95-module computational-biology platform, to an independent external validation: every module family whose internal goldens report verification against live data is re-tested here on held-out public benchmarks its own tests never touched - fresh CRISPR guides mined from a BRCA1 RefSeqGene segment disjoint from the module’s training accessions and scored against the CRISPOR reference implementation of the CFD off-target model; a 500-variant stratified sample of the current ClinVar release; 4,956 single-point mutations from SKEMPI 2.0; the BiGG *E. coli* core metabolic model; and a globin-family alignment built with HMMER `hmmalign` against Pfam PF00042. We report per-module agreement metrics with bootstrap confidence intervals, publish all fixtures with provenance, and document failure modes without suppressing negative results.

1 Introduction

Computational-biology toolkits are typically validated against the same data that informed their development, inflating confidence. Independent, held-out validation against public reference implementations and current database releases is the standard this platform must meet to be usable in laboratory settings. This study asks a single question: does each verified claim in the platform’s own status ledger survive contact with data it has never seen? We answer it module family by module family, with frozen fixtures, pinned random seeds, and metrics reported with 95% bootstrap confidence intervals. All code, fixtures, and results are released in this repository.

2 Related Work and Benchmarks

Reference implementations used here are CRISPOR for guide scoring [2] and its Doench Rule Set 1/CFD models [1], SKEMPI 2.0 for binding free-energy changes [3], BiGG for metabolic models [4], Clustal Omega for alignment [5], and the Tavtigian Bayesian ACMG framework [6]. Pharmacogenomic tables come from the CPIC API; RNA structure references from Rfam seed alignments; splice predictions are compared with SpliceAI.

3 Methods

For each module family we constructed a held-out test disjoint from every fixture used during platform development. Negative results are preserved and reported alongside positives; no test was re-designed after seeing its outcome.

3.1 Data and Provenance

All datasets are public and carry recorded provenance (see `data/README.md`): NCBI ClinVar `variant_summary.txt.gz` (retrieved 2026-09-24; 500-variant stratified reservoir sample from 1,517,551 eligible GRCh38 SNVs with criteria-provided review status); SKEMPI 2.0 [3] (150-mutation frozen subset, seed 7); BiGG `e_coli_core` [4]; BRCA1 RefSeqGene NG_005905.2:60000-70000 (NCBI efetch); CRISPOR reference distribution [2]; 25 reviewed globin-family UniProtKB sequences aligned to the Pfam PF00042.29 HMM with HMMER3 `hmmalign` (pyhmmmer 0.12.3); the EBI Clustal Omega job queued for over 40 minutes and was abandoned.

3.2 Mathematical Formulation

CFD off-target scoring follows Doench et al. 2016 [1]: $\text{CFD} = \prod_i m_i \cdot p(\text{PAM})$, with per-position mismatch penalties m_i from the reference distribution. Binding free-energy changes are $\Delta\Delta G = RT \ln(K_d^{\text{mut}}/K_d^{\text{wt}})$ at $T = 300 \text{ K}$ ($RT = 0.596 \text{ kcal/mol}$). Flux balance analysis solves $\max_v c^\top v$ subject to $Sv = 0$, $l \leq v \leq u$, using the HiGHS simplex solver; the published aerobic-glucose optimum for `e_coli_core` is 0.8739 h^{-1} [4]. Agreement is quantified with Pearson r , Mann-Whitney AUC, and binary-classification metrics; confidence intervals are percentile bootstrap with 1,000 resamples at fixed seed.

3.3 Derivations

(F1) Position weight matrix log-odds. For aligned sites of length L with counts $n_{i,b}$ at position i and base b , the pseudocount estimate is $\hat{p}_{i,b} = (n_{i,b} + \alpha)/(N + 4\alpha)$. The log-odds weight against background q_b is

$$w_{i,b} = \log_2 \frac{\hat{p}_{i,b}}{q_b}, \quad S(x) = \sum_{i=1}^L w_{i,x_i}. \quad (1)$$

$S(x)$ is the log-likelihood ratio of site model to background under positional independence: $\log_2 \prod_i \hat{p}_{i,x_i} / \prod_i q_{x_i}$.

(F2) Normalised score. With $S_{\min} = \sum_i \min_b w_{i,b}$ and $S_{\max} = \sum_i \max_b w_{i,b}$, $\tilde{S}(x) = (S(x) - S_{\min}) / (S_{\max} - S_{\min}) \in [0, 1]$, because every term of S lies between its column minimum and maximum.

(F3) Variant delta. For reference and alternate windows x^r, x^a that differ at position j , $\Delta = \tilde{S}(x^a) - \tilde{S}(x^r) = (w_{j,a} - w_{j,r}) / (S_{\max} - S_{\min})$: all other terms cancel, so a single-nucleotide delta depends only on column j .

(F4) Tract blend. The acceptor score is $A(x) = (1 - \lambda)\tilde{S}(x) + \lambda T(x)$ with $\lambda = 0.25$, where T is the normalised pooled polypyrimidine-tract score. Linearity gives $\Delta A = (1 - \lambda)\Delta\tilde{S} + \lambda\Delta T$.

(F5) AUC as a Mann-Whitney statistic. For scores $\{s_i^+\}_{i=1}^m$ and $\{s_j^-\}_{j=1}^n$,

$$\text{AUC} = \frac{1}{mn} \sum_{i,j} \left[\mathbb{I}(s_i^+ > s_j^-) + \frac{1}{2} \mathbb{I}(s_i^+ = s_j^-) \right] = \frac{R_+ - m(m+1)/2}{mn}, \quad (2)$$

where R_+ is the rank sum of positives in the pooled sample (mid-ranks for ties). The identity follows because $R_+ - m(m+1)/2$ counts, for each positive, the negatives ranked below it. Equivalently, $\text{AUC} = P(s^+ > s^-)$.

(F6) Within-position AUC. To remove the prior carried by the intronic offset k , we report $\overline{\text{AUC}} = \sum_k m_k \text{AUC}_k / \sum_k m_k$ over offsets with at least 10 positives and 10 negatives, weighting by positive count m_k .

(F7) CNN objective. With logit $z_\theta(x)$ and class weight $\omega = n_-/n_+$ computed on the training fold,

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_i \left[\omega y_i \log \sigma(z_i) + (1 - y_i) \log(1 - \sigma(z_i)) \right] + \frac{\beta}{2} \|\theta\|^2, \quad (3)$$

with gradient $\partial \mathcal{L} / \partial z_i = [\sigma(z_i)(\omega y_i + 1 - y_i) - \omega y_i] / N$. Setting $\omega = n_-/n_+$ makes both classes contribute equal total weight.

(F8) Convolutional feature. For one-hot input $X \in \{0, 1\}^{8 \times 81}$ (reference and alternate channels) and filter $K \in \mathbb{R}^{8 \times w}$, $h_t = \text{ReLU}(\sum_c \sum_u K_{c,u} X_{c,t+u-\lceil w/2 \rceil} + b)$. Global max pooling $\max_t h_t$ makes the feature invariant to where the motif sits in the window.

(F9) Binding free energy. $\Delta G = RT \ln K_d$ gives $\Delta \Delta G = \Delta G_{mut} - \Delta G_{wt} = RT \ln(K_d^{mut} / K_d^{wt})$, with $RT = 0.596$ kcal/mol at 300 K.

(F10) FBA duality. The primal $\max c^\top v$ s.t. $Sv = 0$, $l \leq v \leq u$ has the dual $\min u^\top \alpha - l^\top \beta$ s.t. $S^\top \pi + \alpha - \beta = c$, $\alpha, \beta \geq 0$. HiGHS returns primal and dual solutions. Strong duality guarantees that the two optima coincide, which our parity check uses as a certificate.

(F11) Codon Adaptation Index. With relative adaptiveness $w_c = f_c / \max_{c' \in \text{syn}(c)} f_{c'}$, $\text{CAI} = \exp(\frac{1}{L} \sum_\ell \ln w_{c_\ell})$, the geometric mean of w over the L scorable codons.

(F12) ACMG Bayesian posterior. Following Tavtigian et al., $O_P = 350^{\text{pts}/8}$ and $\text{Post} = O_P \pi / ((O_P - 1)\pi + 1)$. With $\pi = 0.1$ and $\text{pts} = 6$: $O_P = 350^{0.75} \approx 80.92$, so $\text{Post} = 8.092/8.992 \approx 0.8999$, matching the module's 0.8999 and the published 0.900.

(F13) Profile-HMM forward recursion. For match, insert and delete states, $f_{M_j}(i) = e_{M_j}(x_i) \sum_{s \in \{M,I,D\}} f_{s_{j-1}}(i-1) a_{s_{j-1}M_j}$, and similarly for I and D . The log-odds score is $\log P(x | \mathcal{M}) - \sum_i \log q_{x_i}$.

(F14) Bootstrap interval. For a statistic $\hat{\theta}$, draw $B = 1000$ resamples with replacement and take the 2.5th and 97.5th percentiles of $\{\hat{\theta}_b^*\}$.

3.4 Derivations for the Second-Round Validations

(F15) Maximum-entropy splice-site score. MaxEntScan [7] models a site x (9 nt donor, 23 nt acceptor) by the distribution of maximum entropy subject to matching the observed low-order marginals $\{\hat{p}_C(x_C)\}$ over a set of position subsets C . By Lagrange duality the solution has Gibbs form

$$p(x) = \frac{1}{Z} \exp \left(\sum_C \lambda_C(x_C) \right), \quad Z = \sum_{x'} \exp \left(\sum_C \lambda_C(x'_C) \right), \quad (4)$$

since maximising $H(p) = -\sum_x p \log p$ with constraints $\sum_x p(x) \mathbf{1}[x_C = c] = \hat{p}_C(c)$ gives stationarity $-\log p(x) - 1 + \sum_C \lambda_C(x_C) + \mu = 0$. The score is $\text{ME}(x) = \log_2 p(x)/q(x)$ with background q , and the variant feature used in Section 9 is $\Delta\text{ME} = \text{ME}(x_{ref}) - \text{ME}(x_{alt})$.

(F16) Logistic model with MaxEnt features. With feature vector

$$z = (S_{ref}, S_{alt}, \Delta S, d, \text{ME}_{ref}/10, \text{ME}_{alt}/10, \Delta\text{ME}/10, e_k),$$

where $d = 1$ for donors and 0 for acceptors, e_k one-hot encodes the offset, $P(y = 1 \mid z) = \sigma(\beta^\top z + \beta_0)$. The log-likelihood $\ell(\beta) = \sum_i y_i \log \sigma_i + (1 - y_i) \log(1 - \sigma_i)$ has gradient $\nabla \ell = \sum_i (y_i - \sigma_i) z_i$ and Hessian $-\sum_i \sigma_i(1 - \sigma_i) z_i z_i^\top$, which is negative semidefinite, so ℓ is concave and the fit has a single optimum.

(F17) Fisher z test for a correlation. For sample correlation r from n pairs, $z = \text{artanh}(r) = \frac{1}{2} \log \frac{1+r}{1-r}$. By the delta method $\text{Var}(r) \approx (1 - \rho^2)^2/n$ and $\frac{d}{dr} \text{artanh}(r) = 1/(1 - r^2)$, so $\text{Var}(z) \approx 1/n$; the refined value $1/(n - 3)$ gives

$$Z = \text{artanh}(r) \sqrt{n - 3} \sim N(0, 1) \text{ under } \rho = 0, \quad p = 2\Phi(-|Z|). \quad (5)$$

We apply it to Spearman ρ (rank correlation), which is an approximation.

(F18) Benjamini-Hochberg adjustment. For ordered $p_{(1)} \leq \dots \leq p_{(m)}$, $q_{(i)} = \min_{j \geq i} \min(1, m p_{(j)}/j)$. Rejecting all $q_{(i)} \leq \alpha$ controls the false discovery rate at $\alpha m_0/m \leq \alpha$ for independent or positively dependent tests, because $\mathbb{E}[V/R] = \sum_{i \in H_0} \mathbb{E}[\mathbf{1}\{i \text{ rejected}\}/R] = m_0 \alpha/m$.

(F19) tRNA adaptation index. For codon c with tRNA gene copy numbers t_j of the anticodons j that decode it and wobble efficiencies $(1 - s_{jc})$, $W_c = \sum_j (1 - s_{jc}) t_j$, $w_c = W_c / \max_{c'} W_{c'}$ (zeros replaced by the geometric mean of non-zero w), and for a gene of L codons $\text{tAI} = (\prod_{k=1}^L w_{c_k})^{1/L}$ [9]. Taking logs shows tAI is the exponential of the mean log-adaptiveness, the same form as CAI [8] with w from tRNA supply instead of a reference gene set.

(F20) Effective number of codons. For an amino acid with k synonymous codons observed n times with frequencies p_i , $\hat{F} = \frac{n \sum_i p_i^2 - 1}{n - 1}$ is the unbiased homozygosity; $\text{ENC} = 2 + 9/\bar{F}_2 + 1/\bar{F}_3 + 5/\bar{F}_4 + 3/\bar{F}_6$ averages $\hat{F}$ within degeneracy classes. Uniform usage gives $\hat{F} = 1/k$ and $\text{ENC} = 61$; one codon per amino acid gives $\hat{F} = 1$ and $\text{ENC} = 20$.

(F21) Shrake-Rupley surface area. Atom a with radius r_a is expanded to $R_a = r_a + r_p$ ($r_p = 1.4 \text{ \AA}$). With N quasi-uniform points u_k on the unit sphere,

$$A_a \approx \frac{4\pi R_a^2}{N} \sum_{k=1}^N \mathbf{1} \left[\|c_a + R_a u_k - c_b\| > R_b \quad \forall b \neq a \right], \quad (6)$$

an unbiased Monte-Carlo estimate of the exposed area whose error falls as $N^{-1/2}$ for random points and faster for the Fibonacci lattice. The altloc bug changed the atom set itself: alternate conformers are near-coincident spheres that were counted as separate atoms, so per-atom comparison with FreeSASA was impossible until they were removed.

(F22) Nearest-neighbour melting temperature. For a duplex with $\Delta H = \Delta H_{init} + \sum_i \Delta H_{i,i+1}$ and ΔS likewise, at equilibrium $\Delta G = \Delta H - T\Delta S = -RT \ln K$. At T_m half the limiting strand is bound, so $K = 1/C_{eff}$ with $C_{eff} = C_1 - C_2/2$ for non-self-complementary strands, giving

$$T_m = \frac{\Delta H}{\Delta S + R \ln C_{eff}}, \quad \Delta S_{[Na^+]} = \Delta S + 0.368 (N - 1) \ln [Na^+]. \quad (7)$$

(F23) Negative-binomial GLM for counts. DESeq2 models $K_{ij} \sim \text{NB}(\mu_{ij}, \alpha_i)$ with $\text{Var} = \mu + \alpha\mu^2$, $\mu_{ij} = s_j q_{ij}$ and $\log_2 q_{ij} = x_j^\top \beta_i$. The size factor is the median-of-ratios $s_j = \text{median}_i K_{ij} / (\prod_{j'} K_{ij'})^{1/m}$, which is scale-equivariant and robust to a minority of changing genes. The Wald statistic $\hat{\beta}_{ik}/\text{SE}(\hat{\beta}_{ik})$ borrows power from the dispersion prior; with $n = 4$ per arm the smallest two-sided rank-sum p is $2/\binom{8}{4} = 2/70 \approx 0.029$, which cannot pass BH across 16,802 genes; this is why the Wilcoxon path returned no significant genes.

(F24) Robinson-Foulds distance. For unrooted trees T_1, T_2 on the same leaves with non-trivial split sets Σ_1, Σ_2 , $d_{RF} = |\Sigma_1 \triangle \Sigma_2|$. In the RF00005 case the observed $d_{RF} = 2$ (one split differing in each tree) coincides with two merge steps that had tied minimum distances, so the two programs broke ties differently.

3.5 Implementation and Testing

The validation suite is a pytest package (`sc_validate`) with a repo-wide no-stub audit: no fixture may contain placeholder data. Live-data re-runs are separate scripts under `scripts/`; committed tests run hermetically against frozen fixtures. CI runs the suite on every push.

4 Results

4.1 CRISPR off-target (CFD) and on-target (Rule Set 1)

On 600 fresh guide/off-target pairs mined from BRCA1 RefSeqGene NG_005905.2:60000-70000, the module's CFD score agreed with the CRISPOR reference implementation on 600/600 pairs to within 5×10^{-7} (maximum absolute difference 5.0×10^{-7}). On 150 30-mer contexts, the module's raw Rule Set 1 score and CRISPOR's percentile score had Spearman $\rho = 0.9995$.

4.2 Metabolic modelling (FBA)

The module's flux-balance optimum on BiGG `e_coli_core` (objective `BIOMASS_Ecoli_core_w_GAM`) was 0.873922 h^{-1} , identical to an independent SciPy/HiGHS solve (0.873922) and to the published 0.8739 h^{-1} .

4.3 Codon adaptation and ACMG Bayesian scoring

Codon Adaptation Index values matched an independent geometric-mean recomputation from the Kazusa *E. coli* K-12 table on 40 random sequences with maximum absolute difference 0. The ACMG Bayesian posterior matched the Tavtigian et al. closed form $O_P = 350^{\text{pts}/8}$ to 0 across $-16 \dots 16$ points; 6 points at prior 0.10 gives posterior 0.900.

4.4 Splice-site PWMs (deepsplice)

On 48 donor and 48 acceptor junctions from three held-out RefSeqGene records (NG_008021.1, NG_007489.1, NG_008245.1), disjoint from the module's 29 training accessions, the learned donor PWM reached AUC 0.997 and the acceptor PWM 0.998 against composition-matched shuffled windows. Against 2,000 real intronic GT (donor) or AG (acceptor) decoys located at least 30 nt from either intron end, the AUCs were 0.961 (donor) and 0.943 (acceptor). On the same decoys the consensus-seed acceptor matrix reached only 0.816, so the learned matrix does generalize.

4.5 Variant interpretation (openclinvar)

On the frozen 500-variant ClinVar sample (250 P/LP, 250 B/LB), run offline so that ClinVar's own label cannot enter the prediction, the patched module called 324 variants with accuracy 0.997 (95% bootstrap CI 0.991-1.000), sensitivity 0.993 and specificity 1.000 (1 false negative, 0 false positives). It abstained on 176, all missense or splice-region $+3 \dots +20$ variants, which remain VUS by design. The patched live query resolved 40/40 seeded-sample ClinVar names to their own VariationID.

4.6 Profile HMM

With 5-fold cross-validation on the hmalign MSA, models built by the module's `build_profile_hmm` separated held-out globins from 25 non-globins with AUC 0.952 on forward log-odds (0.962 length-normalised). The Pfam/HMMER3 reference model reached 1.000 on the same fixtures.

5 Benchmark Comparison

Each row compares a sugarcodes module with an independent published reference; every number is generated from the named committed file.

Module	Reference	Metric	Result (file)
pgx_guidelines	CPIC diplotype tables	agreement	16836/16836 CYP2D6 (sweep_pgx_cplic.json)
bio.codon CAI	Biopython; PaxDb	Pearson; Spearman	0.99969; 0.5797 (sweep_codon_cai.json)
rna_nussinov	ViennaRNA 2.7	bp F1 (tRNA)	0.328 vs 0.714 (sweep_rna.json)
profile_hmm	Pfam PF00042	auc_log_odds_bits	0.952 global, 0.9792 local (profile_hmm_cv*.json)
splice CNN	SpliceAI 1.3.1	AUROC n=757	0.929 vs 0.9775 (headto-head_interim.json)

6 Failure Modes and Negative Results

openclinvar, first run: 500/500 abstentions. ClinVar names such as NM_000038.6(APC):c.1668T>C (p.Asp556=) carry a transcript/gene prefix and a protein-level suffix. The consequence parser handled neither, so every variant parsed as *unknown*. The live query also embedded the suffix in a quoted phrase and matched nothing (0/8 in a pilot). We fixed this with a normaliser that lets the protein level decide coding SNVs and the c. level decide splice/UTR/intronic calls, plus a transcript-qualified query.

deepsplICE, first run: acceptor AUC 0.547. This was a harness defect, not a model defect. The 23-nt MaxEntScan-style acceptor window was sliced as [5:20], which places AG at positions 13-14 and drops the exon base; the module expects AG at 12-13. We reproduced the bug (AUC 0.524 off-frame) and the fix (0.998 at [6:21]), and a regression test now locks the frame.

profile_hmm: residual gap to HMMER. The module originally offered only a global alignment mode. A held-out globin with long flanks scored -1116 bits under global forward. A local forward mode (sugarcode-ai 1d9091a) raises AUC from 0.952 to 0.979 (benchmarks/profile_hmm_cv_local.json); Pfam’s model still reaches 1.000.

mutdock vs SKEMPI: no honest adapter. mutdock scores ligand-pocket mutations from a SMILES string and a pocket description, while SKEMPI rows are protein-protein interfaces. We report naive base-lines instead: the mean predictor has RMSE 1.63 kcal/mol, $|\Delta V|$ has Pearson $r = 0.053$ and hydrophathy loss $r = -0.129$ ($n = 150$).

openclinvar accuracy is not a hard test. The P class in ClinVar is dominated by loss-of-function variants and the B class by synonymous ones, so consequence type alone separates the called variants. Missense variants, the hard case, are still abstained.

7 Module Sweep: Additional Real-Data Results

7.1 RNA secondary structure

Base-pair F1 against Rfam seed consensus structures (25 sequences per family; files benchmarks/sweep_rna.json, sweep_rna_vienna_methods.json).

Family	mean len	Nussinov	MFE	centroid	MEA($\gamma=1$)	MEA($\gamma=4$)
RF00005	73.0	0.328	0.714	0.716	0.703	0.704
RF00001	117.9	0.185	0.531	0.494	0.505	0.504
RF00010	354.5	0.189	0.545	0.594	0.592	0.572

Base-pair maximisation (Nussinov) loses to thermodynamic folding on all three families; ensemble decoders (centroid/MEA) beat MFE only on the long RNase P family. The shipped `rna_nussinov.fold_energy` now delegates to ViennaRNA.

7.2 Codon adaptation index versus protein abundance

On 3489 *E. coli* K-12 genes (NCBI GCF_000005845.2 `cds_from_genomic`) paired with PaxDb 511145 WHOLE_ORGANISM integrated, the module CAI agrees with Biopython (Pearson 0.99969 on 500 genes). Spearman correlation with $\log_{10}$ abundance is 0.4958 with the genome-wide reference table and 0.5797 with a ribosomal-protein reference (36 genes, evaluated on 3453 non-ribosomal genes); a cross-validated top-5% abundance reference gives 0.5741. The ribosomal reference is now shipped as `ecoli_highexpr`. File: `benchmarks/sweep_codon_cai.json`.



Figure 1. Base-pair F1 by folding method (`sweep_rna*.json`).

7.3 Pharmacogenomic phenotype translation versus CPIC

Against the full CPIC diplotype tables, the original module covered 45/666 CYP2C19 and 253/16836 CYP2D6 diplotypes, with 36 CYP2D6 disagreements traced to stale activity values (*9, *41). After vendoring current CPIC tables: 666/666 and 16836/16836. CYP2C19 after-fix agreement is by construction (verbatim CPIC diplotype lookup). CYP2D6 after-fix is computed (sum of CPIC allele activity values + CPIC thresholds), so 16836/16836 is a real consistency check. File: `benchmarks/sweep_pgx_cplic.json`.



Figure 2. CPIC agreement before and after the fix (sweep_pgx_cplic.json).

8 Discovery Track: Splice-Region VUS Reclassification

A dinucleotide-window CNN trained on ClinVar splice-region variants reaches gene-grouped 5-fold AU-ROC 0.935 (logistic 0.912, PWM 0.889); within-position AUROC 0.900. In a head-to-head on $n=757$ variants SpliceAI scores 0.9775 versus 0.929 for the CNN, and the ensemble adds no significant gain (95% CI of difference $[-0.0087, 0.0081]$). We therefore make no state-of-the-art claim. Files: `discovery/splice_region_vus/`.

CNN threshold	TP	FP	sensitivity	FPR	PPV (prior 3%)
0.5	1794	453	0.77	0.0656	0.2663
0.8	1298	146	0.5571	0.02114	0.449
0.9	996	68	0.4275	0.00985	0.5731
0.95	729	33	0.3129	0.00478	0.6694
0.98	435	11	0.1867	0.00159	0.7838



Figure 3. Splice CNN operating points (cnn_calibration.json).

Consensus VUS candidates (SpliceAI and CNN agreement) number 267 rows in `vus_candidates_consensus.tsv`;

none of the top 20 had a matching LOVD record (lovd_top20.json). These are hypotheses for RNA-level testing, not reclassifications.

9 Splice Track II: A 2004 Baseline Beats Our CNN, and What Survives

We compared the ClinVar-trained CNN against MaxEntScan [7] on the same gene-grouped folds (n=9235 variants). MaxEntScan’s ref–alt score difference reached AUROC 0.9635, above the CNN (0.9355), the logistic PWM model (0.9121) and the raw PWM delta (0.8893). The paired bootstrap 95% CI for CNN minus MaxEntScan is [-0.0329, -0.0233], so this is a clear negative result for the CNN. The gap is mostly at acceptors (CNN 0.8794 vs MaxEntScan 0.965, n=3845); at donors the two are close (0.9538 vs 0.9598, n=5390). As a pivot we added the three MaxEntScan features to both models. The logistic model then reached 0.9674 (CI of gain over MaxEntScan [0.0027, 0.0053]), while the CNN with the same features reached 0.9622 (CI [-0.0042, 0.0017], not significant). Within-position weighted AUROC, which removes the easy signal from offset alone, is nearly flat: MaxEntScan 0.949, logistic 0.9492, CNN 0.9472. The gain is real but small. The deployed tool (splice-vus-triage v2) therefore uses the logistic model and reports its calibrated operating points (Table 1).

Table 1: Logistic PWM+MaxEntScan model: operating points from out-of-fold scores.

Threshold	TP	FP	Sensitivity	FPR
0.5	1899	259	0.815	0.03751
0.8	1503	100	0.6451	0.01448
0.9	1230	56	0.5279	0.00811
0.95	950	28	0.4077	0.00406
0.98	692	15	0.297	0.00217



Figure 1: AUROC by model and site type, gene-grouped 5-fold CV.

10 Codon Track II: Cross-Species Test of the Ribosomal Reference Set

Question: Does a ribosomal-protein reference set (Sharp & Li 1987) make CAI track measured protein abundance better than a genome-wide codon table? Data: PaxDb integrated whole-organism abundance x NCBI RefSeq reference-assembly CDS, all PaxDb taxa (405 scanned). Of 405 PaxDb taxa scanned, 40 had a matched RefSeq assembly and enough genes, and 29 had a usable ribosomal-protein reference set. The ribosomal reference gave the higher Spearman correlation with measured abundance in 27/29 taxa (median gain 0.082; Wilcoxon signed-rank $p = 3.3e - 07$, sign test $p = 1.6e - 06$). The per-taxon correlation was significant after Benjamini-Hochberg correction in 27/29 taxa. Exceptions: *Rattus norvegicus*, *Plasmodium falciparum* 3D7. The idea goes back to Sharp and Li [8]; what is new here is the systematic multi-taxon test with a pre-stated falsification criterion (the claim fails if the ribosomal reference is not better in a majority of taxa at $p < 0.01$).



Figure 2: Spearman ρ (CAI vs PaxDb abundance) with the genome-wide versus ribosomal reference, per taxon.

Table 2: Per-taxon results (all taxa with a matched assembly).

Taxid	Organism	Assembly	n	n_{ribo}	ρ_{gen}	ρ_{ribo}	q
10090	<i>Mus musculus</i>	GCF_000001635.27	8089	99	0.0704	0.0775	4.4e-12
10116	<i>Rattus norvegicus</i>	GCF_036323735.1	890	124	0.142	0.1405	3.1e-05
121845	<i>Diaphorina citri</i>	GCF_000475195.1	592	1	0.1458	—	—
1295	<i>Staphylococcus schleiferi</i>	GCF_025558325.1	442	66	0.4583	0.5403	2e-36
1314	<i>Streptococcus pyogenes</i>	GCF_900475035.1	306	66	0.4954	0.6701	9.8e-45
185431	<i>Trypanosoma brucei brucei</i> TREU927	GCF_000002445.2	6302	0	0.3167	—	—
192222	<i>Campylobacter jejuni</i> subsp. <i>jejuni</i> NCTC 11168 = ATCC 700819	GCF_000009085.1	725	66	0.1208	0.2291	5.3e-10
196627	<i>Corynebacterium glutamicum</i> ATCC 13032	GCF_000011325.1	323	72	0.5268	0.6307	7e-40
198214	<i>Shigella flexneri</i> 2a str. 301	GCF_000006925.2	1887	72	0.5511	0.6514	8.1e-249
208964	<i>Pseudomonas aeruginosa</i> PAO1	GCF_000006765.1	4796	72	0.2459	0.4149	2.5e-204
224308	<i>Bacillus subtilis</i> subsp. <i>subtilis</i> str. 168	GCF_000009045.1	3147	64	0.3295	0.4156	3.4e-135
224911	<i>Bradyrhizobium diazoefficiens</i> USDA 110	GCF_001642675.1	494	72	0.4049	0.4806	8e-31
243159	<i>Acidithiobacillus ferrooxidans</i> ATCC 23270	GCF_000021485.1	419	68	0.3378	0.4055	2.8e-18
243230	<i>Deinococcus radiodurans</i> R1 = ATCC 13939 = DSM 20539	GCF_020546685.1	319	66	0.4784	0.5698	2.4e-30

Table 2: Per-taxon results (all taxa with a matched assembly).

Taxid	Organism	Assembly	n	n_{ribo}	ρ_{gen}	ρ_{ribo}	q
246200	Ruegeria pomeroyi DSS-3	GCF_000011965.2	395	70	0.2714	0.46	1.2e-22
2903	Emiliana huxleyi CCMP1516	GCF_000372725.1	484	4	0.4636	–	–
343509	Sodalis glossinidius str. 'morsitans'	GCF_000010085.1	748	72	0.2849	0.4703	1.2e-43
36329	Plasmodium falciparum 3D7	GCF_000002765.6	4946	62	0.056	0.0253	0.078
403677	Phytophthora infestans T30-4	GCF_000142945.1	708	0	0.5278	–	–
4081	Solanum lycopersicum	GCF_036512215.1	426	4	-0.1336	–	–
4097	Nicotiana tabacum	GCF_000715075.1	9304	0	-0.0837	–	–
418985	Tropilaelaps mercedesae	GCA_002081605.1	2213	0	0.2207	–	–
44689	Dictyostelium discoideum AX4	GCF_000004695.1	3183	10	-0.3039	–	–
4932	Saccharomyces cerevisiae S288C	GCF_000146045.2	5587	226	0.4099	0.5924	0
511145	Escherichia coli str. K-12 substr. MG1655	GCF_000005845.2	3486	72	0.4961	0.5809	0
54291	Klebsiella ornithinolytica	GCF_030505655.1	1038	72	0.2053	0.5268	1.2e-78
6239	Caenorhabditis elegans	GCF_000002985.6	11571	0	0.0748	–	–
7227	Drosophila melanogaster	GCF_000001215.4	2251	150	0.0189	0.0813	0.00012
7460	Apis mellifera	GCF_003254395.2	879	4	-0.0468	–	–
8030	Salmo salar	GCF_965601325.2	1779	0	0.1332	–	–
83332	Mycobacterium tuberculosis H37Rv	GCF_000195955.2	3258	76	0.2304	0.2373	6.6e-43
9031	Gallus gallus	GCF_016699485.2	1891	68	0.2134	0.2274	1.6e-23
93061	Staphylococcus aureus subsp. aureus NCTC 8325	GCF_000013425.1	1699	58	0.5221	0.6046	3.6e-182
9544	Macaca mulatta	GCF_049350105.2	3732	84	0.0736	0.0808	1e-06
9606	Homo sapiens	GCF_000001405.40	12343	86	-0.0083	0.0106	0.24
9615	Canis lupus familiaris	GCF_011100685.1	329	81	0.1639	0.1673	0.0025
9823	Sus scrofa	GCF_054392235.1	3010	84	0.0576	0.071	0.00011
9913	Bos taurus	GCF_002263795.3	2694	81	0.0963	0.1177	1.2e-09
99287	Salmonella enterica subsp. enterica serovar Typhimurium str. LT2	GCF_000006945.2	2472	72	0.3119	0.4743	4.4e-144
9986	Oryctolagus cuniculus	GCF_964237555.1	1993	82	0.0999	0.1012	7.4e-06

q : Fisher-z two-sided test of ρ_{ribo} , Benjamini-Hochberg adjusted at full precision (benchmarks/sweep_codon_bh_fdr.json).

10.1 tRNA adaptation index and effective number of codons

Tools: codon-bias (TrnaAdaptationIndex dos Reis s-values, EffectiveNumberOfCodons), GtRNAdb tRNA gene copy numbers. Table 3 gives Spearman ρ with abundance. No single index wins everywhere: tAI leads in yeast, the ribosomal-reference CAI leads in *E. coli*, and –ENC is weakest in all three organisms.

Table 3: Codon indices vs abundance (Spearman ρ).

Organism	n	CAI_{gen}	CAI_{ribo}	tAI	–ENC
Escherichia coli str. K-12 substr. MG1655	3450	0.4961	0.5809	0.5277	0.3836
Bacillus subtilis subsp. subtilis str. 168	3115	0.3295	0.4156	0.4146	0.1735
Saccharomyces cerevisiae S288C	5474	0.4099	0.5924	0.6604	0.4471

11 Docking Track: Real AutoDock Vina Against LP-PDBBind

Source: LP-PDBBind (github.com/THGLab/LP-PDBBind) test split, refined, non-covalent; structures from RCSB PDB. Engine: AutoDock Vina 1.2.7 (exhaustiveness 8, 24 Å box on crystal ligand) + Open Babel 3.1.1 prep, pH 7.4. 73/80 complexes ran; the 7 failures are listed in Table 4. Redocking: top-1 pose RMSD < 2 Å in 52.0% and best-of-9 in 83.6% (RMSD without symmetry correction (conservative)). Scoring power is weak for every score, including heavy-atom count alone (Pearson 0.2445): Vina top pose 0.2854 (95% CI

[0.055, 0.485]), sugarcodes own Vina-like score on the crystal pose 0.2975 (CI [0.027, 0.484]). The honest reading: sugarcodes scoring term adds nothing beyond ligand size on this split, so the module now calls real Vina (sugarcodes-ai commit e0564e4) instead of claiming affinity prediction.



Figure 3: Left: redocking RMSD distribution. Right: Vina top-pose energy vs experimental pK.

Table 4: Docking failures.

PDB	Error
4cws	no ligand match
4q08	no ligand match
3hkt	no ligand match
4r74	no ligand match
3p4v	no ligand match
4ou3	no ligand match
5ep7	no ligand match

Table 5: Per-complex docking results.

PDB	pK	heavy	E_{cryst}	E_{min}	E_{top}	RMSD ₁	RMSD ₉
1cnw	7.72	26	-4.19	-5.37	-7.02	7.26	6.03
1o3l	6.54	24	-7.09	-8.63	-9.54	3.19	1.09
3po6	7.06	19	-4.54	-6.13	-7.14	3.81	0.86
4db7	6.56	18	-6.63	-7.18	-7.41	0.52	0.52
5sz5	7.7	17	-7.33	-7.33	-8.14	0.39	0.39
4mo8	8.0	16	-4.75	-5.50	-6.25	2.29	1.52
4tim	2.16	11	-3.41	-4.13	-5.59	1.66	1.66
1ejn	5.62	25	-8.20	-8.43	-7.93	2.51	2.15
5d3l	6.06	30	-6.22	-6.32	-6.70	8.34	1.64
3b3c	3.98	10	-5.07	-5.76	-6.48	1.42	1.42
3r4n	6.89	23	-6.79	-7.37	-7.66	0.81	0.81
2p4j	8.96	49	-10.13	-10.73	-11.07	1.58	1.58
5sz4	9.7	16	-7.04	-7.33	-7.65	1.79	1.79
4xit	7.27	32	-12.19	-12.80	-13.06	0.96	0.96
5mjn	8.54	18	-6.42	-6.93	-7.76	5.97	1.41
2xjg	7.82	23	-9.04	-9.53	-9.72	0.25	0.25
5ix0	6.28	25	-11.86	-12.14	-12.27	0.27	0.27
5y8y	7.09	25	-6.08	-6.38	-8.16	4.75	0.84
6g38	6.62	19	-7.36	-7.70	-8.12	0.61	0.61
4o04	6.16	26	-11.03	-12.14	-12.36	1.34	1.34
5ewk	4.85	22	-7.91	-7.96	-8.57	0.68	0.68

Table 5: Per-complex docking results.

PDB	pK	heavy	E_{cryst}	E_{min}	E_{top}	RMSD ₁	RMSD ₉
3d8w	8.15	19	-6.96	-7.48	-8.20	1.43	1.43
4rfc	8.24	23	-5.25	-6.44	-8.37	7.44	5.67
3r17	6.41	20	-6.73	-7.38	-7.74	3.57	2.30
4knj	7.95	21	-6.40	-7.32	-8.23	4.81	1.87
6g6t	7.7	19	-7.42	-7.76	-8.31	2.10	0.61
4lxz	6.74	19	-6.37	-6.47	-7.29	6.21	1.95
5o87	6.61	12	-5.46	-6.58	-6.70	0.44	0.44
4j93	4.7	21	-7.50	-7.81	-8.02	0.44	0.44
6hrq	7.09	23	-6.44	-6.94	-7.94	6.23	2.21
4nh7	8.4	33	-11.46	-12.05	-12.52	1.31	1.31
2x7t	6.63	30	-6.30	-7.34	-7.81	6.16	0.77
4fl1	3.28	31	-5.80	-5.98	-7.02	9.18	2.47
3c84	7.85	16	-6.59	-6.98	-7.23	0.39	0.39
4ytic	7.89	26	-9.65	-10.13	-10.15	0.49	0.49
2zdn	6.25	27	-7.86	-8.84	-9.05	1.02	1.02
5f60	7.37	37	-8.55	-8.90	-9.99	4.69	1.63
6hqy	6.44	22	-5.86	-6.97	-7.28	2.03	1.48
5l8c	7.0	35	-8.93	-9.64	-11.31	9.53	0.86
1p1o	5.76	13	-7.26	-7.37	-7.79	0.60	0.60
5htz	8.62	26	-9.05	-9.25	-9.40	0.19	0.19
4oag	6.3	27	-5.33	-7.22	-7.84	9.16	3.85
2j4i	9.0	36	-8.00	-8.56	-8.54	1.38	1.38
5n24	8.92	18	-7.04	-7.44	-7.80	1.68	1.40
1y3x	5.14	33	-9.50	-9.83	-10.15	0.70	0.70
1f5l	5.28	15	-6.27	-7.20	-7.32	2.87	1.51
5aml	9.15	21	-6.79	-6.99	-7.74	5.22	1.14
6u6w	5.33	21	-7.71	-7.69	-8.04	5.13	0.33
5mpn	7.07	23	-6.86	-7.18	-8.21	7.46	1.25
4jfm	5.48	35	-6.54	-7.29	-7.74	0.97	0.97
5ewy	4.3	22	-7.30	-7.87	-8.78	3.50	2.90
4zme	4.3	19	-6.04	-6.27	-6.60	10.33	0.18
4yxi	6.5	11	-5.93	-6.43	-6.62	0.39	0.39
2erz	5.66	21	-7.23	-8.31	-8.65	2.99	0.93
6oe0	8.92	27	-7.95	-8.86	-9.35	0.86	0.86
4b2i	3.0	9	-4.33	-5.04	-5.25	1.97	0.61
4f5y	5.43	46	-10.70	-11.84	-10.97	3.66	3.66
6eq5	4.74	10	-6.31	-6.71	-7.17	2.09	0.51
5g5v	6.7	36	-10.54	-10.73	-11.46	2.75	1.52
5cau	8.8	34	-9.64	-9.84	-9.93	0.65	0.65
1gi1	5.47	19	-6.60	-8.25	-8.38	0.77	0.77
3hkq	5.34	15	-4.38	-5.42	-6.47	3.67	3.04
2xab	9.27	22	-9.46	-9.81	-9.94	0.21	0.21
3hll	8.24	30	-9.62	-10.25	-10.25	0.36	0.36
5f62	8.17	38	-8.66	-9.01	-9.80	4.22	1.46
3ibi	8.59	13	-4.21	-4.73	-4.97	5.02	3.28
3m35	7.22	37	-8.52	-9.07	-9.52	0.90	0.90
1g54	8.82	25	-7.03	-7.63	-8.00	8.95	8.89
4r76	7.85	26	-7.58	-8.26	-8.82	1.62	1.62
1zgi	5.34	32	-9.23	-9.64	-9.98	0.76	0.76
5cap	7.28	29	-8.09	-8.54	-8.89	1.44	1.44
1klj	7.55	37	-8.36	-8.62	-9.29	1.77	1.77
6rfn	6.7	35	-7.10	-8.72	-9.63	6.69	1.10

12 Structure and Primer Utilities Against Reference Tools

SASA. Against FreeSASA 2.x (Shrake-Rupley, 96 points, probe 1.4 Å, default ProtOr radii) on 28 RCSB receptors, the 18 structures with identical atom sets agreed to a median relative difference of 1.2% (max 2.6%) with median per-atom Pearson 0.980. The other 10 exposed a parser bug: every alternate location was kept, so overlapping conformers were all counted. After the fix, atom counts match and per-atom Pearson is 0.9753 (1gi1), 0.9829 (3po6), 0.9813 (4b2i).

Primers. On 5404 20-mers from NM_000546.6, NM_007294.4, NM_000059.4, NM_004333.6, NM_005228.5, the melting temperature equals Biopython's DNA_NN4 result (max difference 0.0). Against primer3 (SantaLucia 1998 table) the mean difference is -0.1738 °C (max 0.9811), a table choice. The stem-length hairpin heuristic has Spearman 0.0402 with primer3's hairpin ΔG , a negative result; the module now calls primer3 when installed.

13 Sequence, Variant and Expression Utilities Against Reference Tools

Phylogenetics. DendroPy 5.1.0 (PhylogeneticDistanceMatrix.nj_tree/upgma_tree, treecompare.symmetric_difference); Biopython DistanceCalculator. RF00001: NJ RF 0, UPGMA RF 0 (patristic Pearson 1.0), distance model k80. RF00005: NJ RF 0, UPGMA RF 2 (patristic Pearson 0.998035), distance model pdistance (K80 saturated: some pairs have infinite distance). RF00010: NJ RF 0, UPGMA RF 0 (patristic Pearson 1.0), distance model k80. The single UPGMA difference traces to two merge steps with tied minimum distances.

ORFs. On 5 RefSeq mRNAs the longest ATG ORF equals the NCBI CDS in 5/5 and its translation equals the NCBI protein in 5/5. Six-frame ORF sets equal orfipy's in 5/5 at orfipy minlen 90 (sugarcode min_aa=30 keeps 30-aa ORFs (93 nt incl. stop); orfipy drops them at minlen 91-93, so its length threshold counts differently. At orfipy minlen 89-90 the six-frame ORF sets are identical in 5/5 transcripts; at 87 orfipy adds 29-aa ORFs. Convention difference, not a bug.)

VCF. 46603 ClinVar GRCh38 records (5 gene regions, remote tabix via pysam 0.24.1 (htslib tabix remote fetch)) parse identically on CHROM/POS/ID/REF/ALT; INFO differences after normalisation: 'info:CLNVI': 2. Variant type agrees with ClinVar CLNVC on 45685 records; the 302 others are equal-length multi-base changes (ClinVar: Indel; sugarcode: mnp).

Differential expression. GEO GSE52778 NCBI raw counts GRCh38.p13; untreated GSM1275862/66/70/74 vs Dex GSM1275863/67/71/75; genes with ≥ 10 counts in ≥ 4 samples. Size factors equal PyDESeq2's (max relative difference 1.8e-09). PyDESeq2 with a cell-line block finds 4451 genes at $\text{padj} < 0.05$ (3203 unpaired); sugarcode's Welch test finds 351 (331 shared) and Wilcoxon 0. Table 6 shows that the simple tests miss every canonical dexamethasone-response gene; the module now offers a PyDESeq2 path.

Table 6: Canonical dexamethasone-response genes, GSE52778.

Gene	log2FC	padj PyDESeq2	padj Welch	padj Wilcoxon
FKBP5	4.03	2.1e-26	0.06	0.15
TSC22D3	3.206	2.1e-19	0.16	0.15
PER1	3.221	2.3e-99	0.05	0.15
DUSP1	2.955	1.3e-140	0.07	0.15
KLF15	4.411	5.0e-114	0.17	0.15
ZBTB16	7.432	1.1e-58	0.12	0.15
CRISPLD2	2.628	1.7e-61	0.13	0.15

14 Differential expression and pathway-level follow-up

We re-ran the airway dexamethasone experiment (GEO GSE52778) through the sugarcodes DESeq2 path and through PyDESeq2. Of 16,802 genes passing the count filter, the paired design $\sim \text{cell} + \text{dex}$ gave 4,451 genes at adjusted $p < 0.05$ and the unpaired design gave 3,203. Size factors agreed to a maximum relative difference of $1.8\text{e-}09$ (benchmarks/sweep_de_pydeseq2.json).

(F25) Negative-binomial GLM. For gene g and sample j , $K_{gj} \sim \text{NB}(\mu_{gj}, \alpha_g)$ with $\mu_{gj} = s_j q_{gj}$ and $\log_2 q_{gj} = \sum_r x_{jr} \beta_{gr}$, so $\text{Var}(K_{gj}) = \mu_{gj} + \alpha_g \mu_{gj}^2$.

(F26) Median-of-ratios size factor. $s_j = \text{median}_{g:K_g > 0} K_{gj} / (\prod_{v=1}^m K_{gv})^{1/m}$.

(F27) Wald statistic. $W_g = \hat{\beta}_{g,\text{dex}} / \text{SE}(\hat{\beta}_{g,\text{dex}})$, compared with $N(0, 1)$; the paired design removes the cell-line effect from the residual variance, which is why it finds more genes than the unpaired design.

14.1 Gene-set over-representation

We took the 536 dex-induced genes (adjusted $p < 0.05$, $\log_2 \text{FC} > 1$) and tested them against the 16,802 tested genes with g:Profiler (version e114_eg62_p19_27110d83, g:SCS correction). 31 terms passed. The top terms are broad (Table 7).

(F28) Hypergeometric tail. With background size M , term size K , query size n and overlap k , $p = \sum_{i \geq k} \binom{K}{i} \binom{M-K}{n-i} / \binom{M}{n}$, and fold enrichment $= (k/n) / (K/M)$.

Source	Term	overlap	size	p_{adj}
GO:BP	multicellular organismal process	228	4764	2.8e-07
GO:BP	signaling	209	4354	4e-06
GO:BP	cell communication	209	4365	5.2e-06
GO:BP	cell adhesion	73	1031	1.5e-05
GO:BP	system process	80	1189	2.1e-05
GO:BP	response to stimulus	261	5987	8.6e-05
GO:BP	signal transduction	192	4028	8.9e-05
GO:BP	anatomical structure development	196	4191	0.00027
GO:BP	cell migration	74	1134	0.00041
KEGG	Cytoskeleton in muscle cells	19	172	0.00042
GO:BP	developmental process	209	4580	0.00047
GO:BP	cell differentiation	152	3041	0.00054

Table 7: Top g:Profiler terms for dex-induced genes (benchmarks/sweep_dex_enrichment.json).

Negative result. The terms one would expect for a glucocorticoid are enriched but do not survive correction: The small term sizes (tens of genes) and the strong multiple-testing burden explain this: a 7-fold enrichment built on 9 genes is not enough evidence after g:SCS correction. We report it as a negative rather than as confirmation of the expected biology.

Term	overlap	size	fold (approx.)	$p_{g.SCS}$
cellular response to glucocorticoid stimulus	9	43	7.16	0.43
response to corticosteroid	10	91	3.76	1
response to glucocorticoid	10	81	4.22	1
cellular response to corticosteroid stimulus	9	48	6.42	1

Table 8: Glucocorticoid terms: large fold change, not significant.

14.2 Reactome pathways

Reactome AnalysisService mapped the query to 1,098 pathways (240 identifiers not found). No pathway reached $FDR < 0.05$; the smallest FDR was 0.11. Reactome uses its whole human gene universe as background, which is a further caveat.

Pathway	found	total	p	FDR
Anchoring fibril formation	5	15	0.00037	0.11
FOXO-mediated transcription	13	110	0.00056	0.11
Laminin interactions	7	31	0.00028	0.11
Metallothioneins bind metals	5	16	0.00049	0.11
Non-integrin membrane-ECM interactions	12	83	0.00015	0.11
Collagen chain trimerization	8	44	0.00044	0.11
FOXO-mediated transcription of oxidative stress, m	8	49	0.00087	0.13
Adipogenesis	15	147	0.00096	0.13
Degradation of the extracellular matrix	15	148	0.001	0.13
Response to metal ions	5	21	0.0016	0.17

Table 9: Top Reactome pathways (benchmarks/sweep_dex_reactome.json).

(F29) Benjamini-Hochberg. For ordered $p_{(1)} \leq \dots \leq p_{(m)}$, $q_{(i)} = \min_{j \geq i} \min\{1, m p_{(j)} / j\}$.

14.3 Protein interaction network

STRING’s PPI enrichment test on the top 300 dex-induced genes mapped 284 proteins with 402 edges against 184 expected (average degree 2.83, local clustering 0.368), reported p below floating-point resolution. Caveat: STRING’s expectation uses a genome-wide background, so genes expressed in airway smooth muscle may be more connected than random genes regardless of dex.

(F30) Edge enrichment ratio. $E = \text{observed edges} / \text{expected edges} = 402 / 184 = 2.18$.

14.4 Network hubs

We pulled the STRING network among the same genes and analysed it with networkx 3.4.2 on STRING API/network (v12): 201 connected nodes, 402 edges, a largest component of 183 nodes. The highest-degree hubs are FOXO1 (21), CEBPD (15), FOXO3 (14), LEP (13), COL4A1 (13), GPX3 (13). Of ten canonical glucocorticoid-receptor targets fixed before the run, 8 are in the network, with mean degree 6.25 against 4.00 overall; a degree permutation test (10,000 draws) gives $p = 0.049$. With eight genes this is weak evidence and we do not claim it as a finding.

(F34) Betweenness centrality. $C_B(v) = \sum_{s \neq v \neq t} \sigma_{st}(v) / \sigma_{st}$, with σ_{st} the number of shortest s - t paths and $\sigma_{st}(v)$ those through v .

(F35) Permutation p -value. $\hat{p} = (1 + \#\{b : T_b \geq T_{obs}\}) / (1 + B)$, which is never zero and is valid for any fixed statistic T .

14.5 Rank-based test: preranked GSEA

Threshold-based over-representation throws away the ranking. We therefore ran preranked GSEA (gseapy 1.3.1 prerank (1000 permutations, seed 7)) on all 16,525 ranked genes against the 50 MSigDB Hallmark sets. 1 set passed $FDR < 0.05$ and 5 passed $FDR < 0.25$ (Table 10). The one significant set, Adipogenesis, is a known glucocorticoid-driven programme; this is a modest, single-set result.

(F32) Enrichment score. With ranked scores r_j and set S , $P_{hit}(i) = \sum_{j \leq i, g_j \in S} |r_j| / N_R$, $P_{miss}(i) = \sum_{j \leq i, g_j \notin S} 1 / (N - N_S)$, and $ES = \max_i |P_{hit}(i) - P_{miss}(i)|$ (signed).

(F33) Normalised ES. $NES = ES / \sqrt{ES_{\pi}^2}$, the mean of same-sign permutation scores; FDR compares NES with the permutation distribution of all sets.

Hallmark set	NES	nominal p	FDR
Adipogenesis	1.70	0	0.038
UV Response Up	1.55	0.00567	0.107
Fatty Acid Metabolism	1.55	0.00559	0.131
TNF-alpha Signaling via NF-kB	1.50	0.00273	0.141
Reactive Oxygen Species Pathway	1.57	0.0249	0.156
Androgen Response	1.39	0.0557	0.256
Xenobiotic Metabolism	1.42	0.0287	0.266
Hypoxia	1.39	0.0294	0.275
UV Response Dn	1.36	0.0591	0.280
mTORC1 Signaling	-1.31	0.0361	0.283

Table 10: Preranked GSEA, Hallmark (benchmarks/sweep_dex_gsea_hallmark.json).

14.6 Independent GO cross-check with goatools

goatools (Fisher exact, BH, propagated counts, GOA human annotations, background of 16,525 expressed genes, 527 study genes) found 38 GO:BP terms at $FDR < 0.05$. Of g:Profiler’s 22 top BP terms, 9 are also significant here. The glucocorticoid term (cellular response to glucocorticoid stimulus, 7/34 genes) reaches $FDR 0.026$ under BH but fails g:Profiler’s g:SCS. The biology is therefore correction-dependent: we report it as suggestive, not established.

15 Pairwise alignment against parasail

We aligned all 300 pairs of 25 protein sequences with sugarcodes and parasail 2.6.1. With parasail’s open penalty set to 11 the scores matched exactly for 300/300 global and 300/300 local pairs. With open penalty

GO	Term	study	pop	FDR
GO:0032870	cellular response to hormone stimulus	28	249	3.4e-05
GO:0009725	response to hormone	38	421	3.4e-05
GO:0007155	cell adhesion	39	513	0.00098
GO:0032502	developmental process	132	2766	0.0011
GO:0007165	signal transduction	130	2703	0.0011
GO:0030334	regulation of cell migration	47	689	0.0011
GO:2000145	regulation of cell motility	48	719	0.0012
GO:0044057	regulation of system process	28	322	0.0015
GO:0071495	cellular response to endogenous stimulus	33	419	0.0022
GO:0040012	regulation of locomotion	48	740	0.0022

Table 11: goatools GO:BP (benchmarks/sweep_dex_goatools.json).

12 only 36 and 65 matched (maximum difference 26 and 5). This fixes the convention: sugarcodes gap.open= −11 is the cost of the first gap residue, i.e. parasail open 11.

(F31) Affine gap cost. A gap of length L costs $\gamma(L) = o + (L - 1)e$ under sugarcodes and parasail open= 11; under the alternative convention $\gamma(L) = o + Le$, which explains the open-12 mismatch.



Figure 4: Left: Reactome FDRs for the top pathways; the red line is 0.05. Right: STRING observed versus expected edges.

16 Sequence properties, restriction digests and file formats

This section checks four bio/ libraries against Biopython 1.88 ProtParam + Restriction and gffutils. Three pass exactly. One, the restriction-site finder, had a real bug that we fixed.

16.1 Protein physicochemical properties

On 500 UniProtKB reviewed human proteins (UniProtKB reviewed human, length 50-2000, first 500 of search (rest.uniprot.org, 2026-09-25)), every property matches ProtParam (Table 12).

property	n	within tol.	max abs diff	Pearson
mw	500	500 (rel 1e-06)	0	1.0000000000
gravy	500	500 (rel 1e-06)	0	1.0000000000
aromaticity	500	500 (rel 1e-06)	2.8e-17	1.0000000000
instability	500	500 (rel 1e-06)	0	1.0000000000
pI	500	500 (abs 0.01)	5.9e-05	0.9999999999
ext_reduced	500	500 (rel 1e-06)	0	1.0000000000
ext_oxidized	500	500 (rel 1e-06)	0	1.0000000000

Table 12: bio.proteinprops vs Biopython ProtParam (benchmarks/sweep_proteinprops_restriction.json).

(F36) Average molecular weight. $M = \sum_a n_a m_a - (L - 1) m_{\text{H}_2\text{O}}$, with m_a the average residue mass (free amino acid) and one water lost per peptide bond.

(F37) Net charge. $Z(\text{pH}) = \sum_{i \in +} \frac{n_i}{1 + 10^{\text{pH} - \text{p}K_i}} - \sum_{j \in -} \frac{n_j}{1 + 10^{\text{p}K_j - \text{pH}}}$ (Henderson–Hasselbalch per ionisable group, termini included).

(F38) Isoelectric point. pI is the root of $Z(\text{pH}) = 0$; Z is strictly decreasing in pH, so bisection on $[0, 14]$ converges, and the residual 6×10^{-5} difference is the bisection tolerance.

(F39) Instability index. $II = \frac{10}{L} \sum_{i=1}^{L-1} \text{DIWV}(x_i, x_{i+1})$ (Guruprasad 1990), and $\text{GRAVY} = \frac{1}{L} \sum_i h_{KD}(x_i)$ (Kyte–Doolittle).

(F40) Extinction coefficient at 280 nm. $\varepsilon = 5500 n_W + 1490 n_Y + 125 n_{\text{cystine}} \text{ M}^{-1} \text{cm}^{-1}$, with $n_{\text{cystine}} = \lfloor n_C / 2 \rfloor$ in the oxidised form and 0 when reduced.

16.2 Restriction sites: a bug in degenerate recognition sequences

We digested pBR322 (J01749.1, circular, 4,361 bp) and phage lambda (NC_001416.1, linear, 48,502 bp) with all 610 enzymes and compared cut positions with Biopython Restriction. Before the fix (sugarcode-ai 699fee), 533 and 529 enzymes agreed. Two causes explained the disagreements. First, the reverse-strand pattern for a non-palindromic-looking site was built with a plain DNA reverse complement that leaves IUPAC codes unchanged. AccI (GT[^]MKAC) is in fact palindromic under IUPAC complementation (M $\leftrightarrow$ K), but the plain reverse complement GTKMAC was searched as a second pattern, which matches GTGAAC and GTTCAC, neither an AccI site. On pBR322 this gave 5 AccI sites where Biopython finds 2. Second, on linear DNA the finder reported Type IIS cuts that fall outside the molecule. After the fix (sugarcode-ai fe2185d), 608 and 608 enzymes agree. The remaining two, LpnPI, MspJI, are modification-dependent enzymes that do not cut unmethylated DNA; we leave them open.

(F41) IUPAC complement. The map c : A $\leftrightarrow$ T, C $\leftrightarrow$ G, R $\leftrightarrow$ Y, K $\leftrightarrow$ M, B $\leftrightarrow$ V, D $\leftrightarrow$ H, with S, W, N fixed, is the unique involution on the 15 codes that complements every base in the set a code denotes. A site s is palindromic iff $s = \overline{c(s)}$ (reversed), which for AccI holds only under c , not under the four-letter map.

16.3 Stockholm alignments

bio.stockholm parsed 15 Rfam seed alignments (4,471 sequences). 15/15 match Biopython on sequence identifiers and order, aligned sequences and SS_cons, and survive a write–parse round trip unchanged.

file	AC	seqs	width	all checks
RF00001.sto	RF00001	712	230	yes
RF00005.sto	RF00005	954	118	yes
RF00010.sto	RF00010	458	996	yes
RF00004.sto	RF00004	208	278	yes
RF00015.sto	RF00015	177	261	yes
RF00017.sto	RF00017	91	412	yes
RF00023.sto	RF00023	477	954	yes
RF00050.sto	RF00050	146	221	yes
RF00059.sto	RF00059	115	211	yes
RF00162.sto	RF00162	457	239	yes
RF00167.sto	RF00167	133	113	yes
RF00168.sto	RF00168	47	274	yes
RF00174.sto	RF00174	434	461	yes
RF00234.sto	RF00234	18	236	yes
RF00504.sto	RF00504	44	215	yes

Table 13: bio.stockholm vs Biopython AlignIO (benchmarks/sweep_stockholm_biopython.json).

(F42) Pairwise identity. $PID(x, y) = \#\{i : x_i = y_i, x_i, y_i \notin \{-, .\}\} / \#\{i : x_i, y_i \notin \{-, .\}\}$, case-insensitive; our implementation equals an independent count on every file.

16.4 GFF3 annotation

On the RefSeq E. coli K-12 annotation (9,523 records), bio.gff and gffutils 0.14 (sqlite, in-memory) agree on every record (9,523/9,523) and on the children of 300/300 genes, and write–parse is lossless. A first comparison showed two child mismatches; both were CDS features split over two lines with a shared ID, which gffutils renames X_1. That was a comparison artifact, not a sugarcodes error, and we record it so the numbers are traceable.

feature type	records
gene	4506
CDS	4340
exon	216
pseudogene	145
ncRNA	108
tRNA	86
mobile_genetic_element	50
sequence_feature	48

Table 14: Feature types in GCF_000005845.2 (benchmarks/sweep_gff_gffutils.json).

16.5 Newick trees

bio.newick parsed 6 Open Tree of Life synthetic subtrees with 3,655 leaves in total, including quoted labels with spaces and parentheses and empty leaves left by the height limit. Leaf sets equal Biopython Bio.Phylo and DendroPy, internal-node counts and internal labels equal Bio.Phylo, and write–parse is lossless in 6/6 trees. These trees have no branch lengths, so this checks topology and labels only.

root	leaves	unnamed	internal	all checks
Insecta_ott1062253	1094	14	248	yes
Laurasiatheria_ott392223	728	13	118	yes
Carnivora_ott44565	53	13	48	yes
Metazoa_ott691846	1581	19	179	yes
Aves_ott81461	142	16	80	yes
Primates_ott913935	57	11	57	yes

Table 15: bio.newick vs Bio.Phylo/DendroPy (benchmarks/sweep_newick_biophylo.json).

(F43) Tree size identity. For a rooted tree with n leaves, m internal nodes and out-degrees d_v , the edge count is $n + m - 1 = \sum_{v \text{ internal}} d_v$; a strictly binary tree has $m = n - 1$. The Carnivora subtree has 53 leaves and 48 internal nodes, so it is far from binary, which exercises the parser on polytomies.

16.6 SAM records and a second bug

bio.sam parsed 10,943 real reads (1000 Genomes phase 3 NA12878 chrom20 low-coverage BAM, 20:1000000-1200000). Name, flag, reference, position, mapping quality, CIGAR operations, mate fields, sequence, qualities and typed tags equal pysam for all 10,943 reads. The derived reference end did not: 10,905/10,943 before the fix. The 38 failures were unmapped mates (flag 0x4) that bwa places at the mapped mate’s position and gives a CIGAR. The SAM specification says those fields are not an alignment, and htslib reports no end; sugarcodes computed one. After sugarcodes-ai 87cf78e all 10,943 agree.

(F44) Reference span of a CIGAR. $\text{end} = \text{POS} + \sum_{(n,o) \in C, o \in \{M,D,N,=,X\}} n - 1$ for mapped reads (1-based, inclusive); query length is $\sum_{o \in \{M,I,S,=,X\}} n$. For flag & 4 $\neq 0$ the end is undefined.

16.7 FASTQ reads

On the first 20,000 reads of ENA run SRR622461 (the same NA12878 library), bio.fastq equals Biopython on identifiers (20,000), sequences (20,000) and per-base Phred scores (20,000), detects offset 33, and round-trips losslessly. GC was 0.4252 against 0.4252 (Biopython, ambiguous bases removed) and 0.422 (ambiguous bases kept in the denominator). The difference is a convention: sugarcodes divides by A+C+G+T. We document it rather than change it.

(F45) Phred score. $Q = -10 \log_{10} P_{err}$, encoded as ASCII $Q + 33$; the mean read quality reported is the arithmetic mean of Q , not $-10 \log_{10}$ of the mean error probability, which would be lower for reads with a few bad bases.

16.8 PDB and mmCIF structures: a third bug

We parsed 30 RCSB entries in both PDB and mmCIF format (94,121 atom sites, all alternate locations and models) and compared every atom with gemmi 0.7.5. The PDB-format parser was identical in 30/30 entries. The mmCIF parser failed on all 30 real files (error: loop_ row count 3 is not a multiple of 4 columns (all 30 files)). Two causes: POSIX shell-style splitting broke atom names containing a prime ($O5'$, $C1'$), and loops were closed early when a row contained an unquoted item name as a value, which RCSB files do in the revision-history tables. A CIF 1.1 tokenizer (a quote closes only when followed by whitespace) and closing loops only at row boundaries fixed both (sugarcode-ai fc1fd1e); mmCIF is now identical to gemmi in 30/30 entries. Unit tests with toy CIF strings had passed throughout. Only real files exposed the bug.

(F46) Loop row invariant. A loop with c columns and t value tokens is well formed iff $t \equiv 0 \pmod{c}$; closing the loop at the first reserved-looking token is only safe when the running count $t \bmod c = 0$, which is the rule we now apply.

16.9 Pileup

Using the same reads, bio.pileup and pysam 0.24.1 pileup (stepper nofilter, flag_filter=4, no quality filters, overlaps kept) agree on base counts at 197,437/197,437 positions and on deletion counts at 197,437/197,437. sugarcode also writes deletions as ‘*’ in the base counts, following samtools mpileup; the comparison maps these to the deletion count.

(F49) Column depth. $D(p) = \sum_r \mathbb{I}[p \in [\text{POS}_r, \text{end}_r] \wedge \text{op}_r(p) \in \{M, =, X, D\}]$, split into base-covering reads ($M, =, X$) and deletions (D); reference skips (N) and clipped bases do not count.

16.10 Transcription-factor motifs

We read 10 JASPAR 2024 profiles and scanned BRCA1 (10 kb) and phage lambda (48.5 kb) on both strands, comparing with Biopython 1.88 Bio.motifs (jaspar reader, pseudocount 0.5, uniform background, log2). Log-odds matrices agree to 2×10^{-15} , per-window scores to 4×10^{-6} (Biopython stores scores in single precision), and the hit sets at 80% of the score range are identical for every motif and sequence.

motif	length	hits BRCA1	hits lambda
MA0003.4 TFAP2A	14	27	61
MA0079.3 SP1	11	235	599
MA0080.4 SPI1	14	16	61
MA0099.3 FOS::JUN	10	77	286
MA0105.4 NFkB1	13	29	52
MA0106.3 TP53	18	8	21
MA0108.2 TBP	15	154	530
MA0112.3 ESR1	17	11	18
MA0113.3 NR3C1	17	2	36
MA0139.1 CTCF	19	13	40

Table 16: Motif hits at 80% of range, identical to Biopython (benchmarks/sweep_motif_biopython.json).

(F50) Log-odds score and relative threshold. $S(w) = \sum_{i=1}^L \log_2 \frac{(c_{i,w_i} + \alpha)/(N_i + 4\alpha)}{q_{w_i}}$ with pseudocount $\alpha = 0.5$ and background $q = 1/4$; a window is a hit when $S(w) \geq S_{\min} + f(S_{\max} - S_{\min})$, $f = 0.8$, where $S_{\max} = \sum_i \max_b$ and $S_{\min} = \sum_i \min_b$ of the log-odds matrix.

16.11 BED intervals: a fourth bug

We fetched two hg38 tracks from the UCSC REST API: CpG islands on chr20 (847 intervals) and RepeatMasker on chr20:0–5 Mb (11,699 intervals), wrote them as BED and compared bio.bed with bioframe 0.8.0 (merge, overlap). Parsing and 600 random overlap queries agreed exactly (300/300 and 300/300). merge_intervals did not: its docstring promised bedtools-style merging, yet it refused to join book-ended intervals. RepeatMasker has 3,112 book-ended pairs, so sugarcodes returned 11,364 merged intervals where bedtools/bioframe return 8,251. CpG islands have no book-ended pairs and hid the bug (847 = 847). We added a bedtools-compatible min_dist argument (default 0; -1 for overlap-only) and a matching CLI flag; after the fix both tracks match bioframe exactly (8,251 = 8,251).

(F51) Merge rule. For intervals sorted by start with running end e^* , interval $[s, e)$ joins the current block iff $s \leq e^* + d$; $d = 0$ joins book-ended intervals ($s = e^*$) and $d = -1$ requires at least one shared base ($s < e^*$).

16.12 GenBank flat files: a fifth bug

We fetched 25 GenBank records one by one from NCBI nuccore (phage, viral, mitochondrial, bacterial, RefSeqGene and 13 human RefSeq mRNAs) and compared bio.genbank with Biopython 1.88 SeqIO genbank on 1,867 features. Sequences and feature counts already agreed, but only 1,829/1,867 feature locations and 5/302 CDS /translation qualifiers matched. The parser appended every continuation line of a multi-line qualifier to the feature location instead of to the qualifier: /translation kept its first line only, and when a continuation began with digits the location itself was corrupted (E. coli thrA CDS: 337..2799 became 337..27991). Partial locations (;1..;200) and single bases produced no spans at all. We rewrote qualifier continuation (INSDC joining: no spaces for /translation, one space otherwise, doubled quotes unescaped), added a location parser for join/order, complement, partial ends, between-base sites and remote parts, and kept all IUPAC letters in ORIGIN so coordinates cannot shift. After the fix: spans 1,867/1,867, strands 1,867/1,867, /translation 302/302, CDS extraction 302/302. The fix exposed a downstream case: KCNQ1's record NG_016178.2 carries only a partial CDS, complement(;42918..;43038). The old parser read no spans and bio.splice reported 'no CDS spans'; with spans read, the junction map came back 'ok' but empty. We made it fail loud with status 'partial CDS in record'. The splice-region discovery pipeline reads hg38 directly and does not use this parser, so its results are unchanged.

(F52) Qualifier continuation. For a qualifier whose value spans lines $\ell_1, \dots, \ell_m$, $v = \ell_1 \oplus_{\sigma} \ell_2 \oplus_{\sigma} \dots \oplus_{\sigma} \ell_m$ with separator $\sigma = \emptyset$ for /translation and $\sigma = \text{one space}$ otherwise; a line starting with / opens a new qualifier only when the running count of quote characters in v is even.

16.13 Core sequence operations

On the same 25 GenBank records, bio.sequence.translate reproduced the /translation qualifier for 285/285 standard-code CDSs (17 CDSs with other genetic codes, codon_start $\neq$ 1, transl_except or ribosomal slippage were set aside) and matched Bio.Seq.translate on all 285. Reverse complements agreed on 25/25 whole

records, IUPAC motif search on 192/192 (motif, record) pairs, Wallace T_m on 500/500 lambda oligos (8–13 nt), and ORF calls (≥ 100 aa, both strands) on 22/22 records (239 = 239 ORFs) against an independent Biopython frame scan. GC fraction agreed to 10^{-12} on 24/25; the exception is the human mitochondrial genome, whose single N sugarcode leaves out of the denominator. Protein masses on 500 UniProt proteins differ from ProteinAnalysis by at most $5.0e-05$ (relative), from two-decimal residue masses. Reading the code for this comparison turned up a latent bug the data could not: `clean_dna` silently dropped IUPAC ambiguity letters, so `ATGRCTTAA` translated as `ML` (frame shifted) instead of `MX`. No CDS in this set carries an ambiguity code, so the real-data numbers are identical before and after; ambiguity letters now become N (sugarcode 143d1c4), with a regression test. We report it as a code-review finding, not a data finding.

(F53) Frame preservation. Cleaning must be a letter-wise map $c : \Sigma \rightarrow \{A, C, G, T, N\} \cup \{\epsilon\}$ with $c(x) = \epsilon$ only for non-letters; then codon k of the cleaned sequence is $c(s_{3k})c(s_{3k+1})c(s_{3k+2})$ and every downstream codon keeps its frame.

16.14 k-mer sets and sketches: exact agreement, a biased estimator

We fetched six coronavirus genomes (SARS-CoV-2, bat RaTG13, pangolin GX-P4L, SARS-CoV Tor2, bat ZC45, MERS) and compared `bio.kmer` with `sourmash` 4.8.14 at $k = 21$. With `sourmash` keeping every hash (scaled=1), distinct canonical k-mer counts agreed for 6/6 genomes, and Jaccard and containment agreed exactly on all 15 pairs (maximum difference 0.0). The sketch estimator is a negative result. `bio.kmer`'s minimizer sketch (window $w = 100$, about 600 k-mers per genome) under-estimated Jaccard: for SARS-CoV-2 vs RaTG13 the exact value is 0.321, the minimizer estimate 0.236 and `sourmash` `FracMinHash` (scaled=100, about 300 hashes) 0.312. Across the 15 pairs the mean absolute error was 0.0121 for minimizers and 0.0044 for `FracMinHash`, with half the sketch size. Jaccard over minimizer sets is known to be biased because sequence changes shift which k-mer wins each window; `FracMinHash` keeps a k-mer if and only if its hash falls below a fixed threshold, so the kept subset is a uniform sample of the union. The module's docstring already labels the values as estimates; we record the bias and do not recommend minimizer Jaccard for distance estimation.

(F54) FracMinHash estimator. With hash h and threshold H/s , $\text{FMH}_s(A) = \{a \in A : h(a) < H/s\}$ and $\hat{f} = |\text{FMH}_s(A) \cap \text{FMH}_s(B)| / |\text{FMH}_s(A \cup B)|$, which is (nearly) unbiased because inclusion depends on the k-mer alone; the Mash distance is $D = -\frac{1}{k} \ln \frac{2J}{1+J}$.

16.15 Population-genetics statistics on 1000 Genomes

We pulled 1000 Genomes phase 3 genotype counts for 22 well-studied SNPs (APOE, LCT, SLC24A5, HERC2, ALDH2, ABCC11, ADH1B, EDAR, F5, HFE, MTHFR, COMT and others) from the Ensembl GRCh37 REST service, one record per rsID, giving 110 (SNP, super-population) genotype tables. `bio.gstats.hwe_exact` matched an independent full enumeration of the exact HWE distribution on 110/110 tables (maximum relative error $1.1e-12$); 15 tables have $p < 0.05$, as expected for a set that includes SNPs with strong population structure inside super-populations. Treating EUR as 'cases' and EAS as 'controls' for 20 SNPs, allelic chi-square (with and without Yates), genotypic chi-square and Woolf odds ratios matched `scipy` and `statsmodels` to $\leq 1e-13$ relative error, and Benjamini-Hochberg and Bonferroni adjustments matched `statsmodels` exactly (maximum difference $2e-19$). No bug was found in this module.

SNP	alleles	EUR (AA/AB/BB)	EAS (AA/AB/BB)	allelic χ^2	OR
rs1426654	A/G	500/3/0	0/12/492	1954.6	2.77e+04
rs16891982	C/G	4/54/445	498/6/0	1756.6	0.000393
rs3827760	A/G	492/11/0	12/104/388	1516.9	622
rs1229984	T/C	1/27/475	244/215/45	972.8	0.0129
rs12913832	A/G	90/186/227	502/2/0	932.6	0.00114
rs17822931	C/T	378/113/12	36/150/318	840.0	22.5
rs4988235	G/A	154/187/162	504/0/0	686.1	0.00048
rs671	G/N	503/0/0	349/135/20	191.3	424
rs53576	A/G	55/243/205	212/231/61	179.9	0.291
rs2231142	G/T	414/83/6	251/213/40	124.7	3.93
rs1799945	C/G	348/137/18	478/23/3	114.4	0.143
rs4680	G/A	133/237/133	264/198/42	102.7	0.388
rs1042713	G/A	183/252/68	96/263/145	53.7	1.94
rs1800562	G/A	461/41/1	504/0/0	44.0	0.011
rs429358	T/C	358/134/11	421/79/4	22.4	0.515
rs1801133	G/A	204/231/68	259/192/53	10.9	0.731
rs7412	C/T	441/61/1	407/93/4	9.5	1.67
rs10427255	C/T	131/236/136	94/257/153	5.8	1.24
rs1815739	T/C	90/257/156	98/251/155	0.2	0.964
rs762551	C/A	58/206/239	55/220/229	0.1	0.967

Table 17: EUR vs EAS allele-frequency contrasts from 1000 Genomes phase 3, computed by bio.gstats and matching scipy/statsmodels (benchmarks/sweep_gstats_scipy.json).

(F55) Exact HWE distribution. For n individuals with n_A copies of allele A and $n_B = 2n - n_A$, the probability of h heterozygotes is $P(h) = \frac{n!}{n_{AA}!h!n_{BB}!} \frac{2^h n_A!n_B!}{(2n)!}$ with $n_{AA} = (n_A - h)/2$, $n_{BB} = (n_B - h)/2$; the two-sided p-value is $\sum_{h:P(h) \leq P(h_{obs})} P(h)$.

16.16 UniProt lookup: a wrong protein and a cache that never hit

bio.uniprot.search maps a gene symbol to its UniProt entry, and several modules (binding sites, mutation docking) build on it. We checked it against HGNC’s curated symbol-to-UniProt mapping for 60 human genes, fetching each HGNC record separately. Before the fix 59/60 matched. The miss was CDKN2A: the query gene:CDKN2A also matches CDKN2A-AS1, and taking the top hit returned Q9UH64 (the antisense-locus protein) instead of P42771 (p16-INK4a). We switched to gene_exact and prefer the entry whose primary gene name equals the symbol; after the fix 60/60 match and all 60 returned sequences have the length of the entry fetched by accession. Reading the connector turned up a second bug. The on-disk cache named each file by abs(hash(url)); Python salts string hashes per process, so a new process never found its own cache and offline=True always failed. The same pattern was in the ChEMBL, gnomAD and structure connectors. All four now use SHA-256, and a new process running offline reads TP53 and CDKN2A from cache (P04637, P42771).

(F56) Stable cache key. A cache key must be a pure function of the request, $k(u) = \text{SHA256}(u)$; a per-process salted hash $k_s(u)$ gives $\Pr[k_{s_1}(u) = k_{s_2}(u)] \approx 2^{-64}$ for independent salts, so the hit rate across processes is effectively zero.

16.17 NCBI Gene lookup: Huntington resolved to the serotonin transporter

bio.entrez.gene_id resolves a symbol to an NCBI Gene ID and feeds the gene-analysis module. Against HGNC’s curated Gene IDs for the same 60 symbols it scored 58/60. HTT returned 6532, which is SLC6A4 (the serotonin transporter, alias HTT), not huntingtin (3064); TTN returned 7276, which is TTR (transthyretin, alias TTN), not titin (7273). The query field [sym] also matches aliases and the search took the top hit. The function now fetches up to ten hits and returns the one whose official symbol equals the query; after the fix 60/60 match (sugarcode 93d5b12). For a user asking about Huntington disease or titin cardiomyopathy the old answer would have been about the wrong gene.

(F57) Symbol resolution rule. Given candidate IDs $u_1, \dots, u_r$ from an alias-aware search, return $u^* = u_{\min\{i: \text{sym}(u_i)=q\}}$ if that set is non-empty, else u_1 ; an alias match is used only when no official-symbol match exists.

17 Adding evolutionary conservation to the splice-region model

The best splice-region model so far is a logistic regression on PWM and MaxEntScan features. Neither feature sees conservation. We fetched phyloP100way at each variant position from the UCSC Genome Browser REST API for a stratified subsample of 1,501 of the 9,235 modelled variants (379 pathogenic, 1,054 genes, 0 missing). The subsample keeps the class-by-site proportions of the full set; the API rate limit (one request per second) is why we did not fetch all 9,235. Median phyloP is 3.214 at pathogenic and 0.191 at benign positions. On its own, phyloP reaches AUROC 0.8377 (0.9108 at donors, 0.6966 at acceptors). Added to logit+MaxEnt under gene-grouped 5-fold cross-validation repeated over 20 seeds, AUROC rises from 0.963 to 0.9678. A paired bootstrap on the averaged out-of-fold predictions gives a 95% interval for the gain of [+0.0008, +0.0091], which excludes zero. The gain is concentrated at donor-region positions

site	n	phyloP alone	logit+MaxEnt	+ phyloP
donor	876	0.9108	0.9557	0.9634
acceptor	625	0.6966	0.9702	0.9723
all	1501	0.8377	0.9630	0.9678

Table 18: AUROC with and without phyloP (discovery/splice_region_vus/phylop_lift.json). Per-site values use the seed-averaged predictions.

(+3 to +6), where the intronic consensus is short and conservation carries information the PWM misses. At acceptors (the polypyrimidine tract, −3 to −14), MaxEntScan already scores well and conservation adds little.

Caveat. ClinVar submitters may cite conservation as supporting evidence (ACMG criteria PP3/BP4). Some of the gain may therefore be label circularity rather than new biology. We report the gain as modest and conditional on that caveat.

(F47) Model with conservation. $\text{logit}P(\text{path} \mid x) = \beta_0 + \beta^\top \mathbf{z}(x) + \gamma \phi(x) + \delta \phi(x) \mathbb{I}[\text{donor}]$, with $\mathbf{z}$ the standardised PWM/MaxEnt/position features and ϕ the phyloP score; the interaction lets conservation matter differently at donors.

(F48) Paired bootstrap. For resamples $b = 1..B$ of variant indices I_b , $\Delta_b = \text{AUC}(y_{I_b}, \hat{p}_{I_b}^+) - \text{AUC}(y_{I_b}, \hat{p}_{I_b}^0)$; the interval is the 2.5th and 97.5th percentile of $\{\Delta_b\}$. Pairing on the same resample removes the shared sampling noise of the two models.

18 Discussion

Validating each module against an independent reference exposed three kinds of result: implementations that match their reference exactly (CAI vs Biopython, CYP2D6 translation vs CPIC after the fix), implementations whose defaults were stale or weak (CPIC activity values, genome-wide CAI reference, Nussinov folding), and places where a strong published tool beats our model (SpliceAI on splice-region variants). Each fix is committed with the result file that motivated it. The CNN is useful as a calibrated, cheap second opinion at high thresholds, not as a replacement for SpliceAI.

19 Conclusion

The sweep so far corrected two clinically relevant defaults (CPIC activity values, clopidogrel handling of “Likely” phenotypes), improved CAI correlation with measured abundance, and replaced base-pair maximisation with thermodynamic folding. The accession-level dataset target (120) and the formula target (10) are met; the external-tool target (40) is met (manifests/tools.tsv). The manifests in the appendix list exactly what was used.

20 Module inventory and validation coverage

sugarcode-ai contains 95 application modules and 32 bio/ libraries. 38 of these 127 have been checked against an independent reference on real data in this work (11 modules, 27 libraries). The rest are not yet validated. That does not mean they are wrong, only that no claim about them is made here. Validation status is a curated list tied to committed result files (manifests/module_inventory.tsv, built by sweep/module_inventory.py), not a keyword match.

kind	name	py lines	status
bio	align	164	validated
bio	bed	180	validated
bio	codon	170	validated
bio	de	218	validated
bio	entrez	196	validated
bio	fastq	183	validated
bio	genbank	103	validated
bio	gff	237	validated
bio	gstats	194	validated
bio	kmer	261	validated
bio	motif	198	validated
bio	newick	248	validated
bio	orf	189	validated
bio	pdb	346	validated
bio	phylo	232	validated
bio	pileup	148	validated
bio	primer	278	validated
bio	proteinprops	261	validated
bio	pwm	63	validated
bio	restriction	150	validated
bio	maseq	201	validated

kind	name	py lines	status
bio	sam	221	validated
bio	sequence	174	validated
bio	stockholm	260	validated
bio	structures	267	validated
bio	uniprot	106	validated
bio	vcf	263	validated
module	chem_descriptors	923	validated
module	chem_similarity	234	validated
module	codon_opt	289	validated
module	crisprscan_score	178	validated
module	deepsplice	751	validated
module	docking_studio	535	validated
module	mit_offtarget	154	validated
module	neohunter	358	validated
module	pgx_guidelines	287	validated
module	profile_hmm	747	validated
module	rna_nussinov	387	validated
bio	chembl	176	not yet
bio	fasta	55	not yet
bio	gnomad	96	not yet
bio	reactome	63	not yet
bio	splice	682	not yet
module	acmg_bayesian	457	not yet
module	alpha_fold_ui	404	not yet
module	bio_copilot	245	not yet
module	bio_material	114	not yet
module	bio_switch	511	not yet
module	biofactory_1_a	102	not yet
module	biogpt_lit	249	not yet
module	bioimage_ai	146	not yet
module	bioplayground	98	not yet
module	bioprint_pro	143	not yet
module	biosimvr	200	not yet
module	car_t_designer	104	not yet
module	cell_free_opt	107	not yet
module	cell_twin	170	not yet
module	cellfatenet	199	not yet
module	cellpainter	289	not yet
module	cellpainter_4d	164	not yet
module	cf_d_offtarget	157	not yet
module	chemgpt_engine	305	not yet
module	crispr_cargo	156	not yet
module	crispr_muse	278	not yet
module	crispr_opt	648	not yet
module	crisprater	97	not yet
module	dark_genome	202	not yet
module	dna_to_code	111	not yet
module	dtt_bench	196	not yet
module	ecosystem	175	not yet
module	enterprise_bio	260	not yet
module	epi_edit	178	not yet
module	evidence_mining	418	not yet
module	evofold_4d	236	not yet
module	fate_predictor	131	not yet
module	gene_analysis	262	not yet
module	gene_explorer	140	not yet
module	gene_tx_opt	149	not yet
module	genomegpt	279	not yet
module	infinite_diagnosis	130	not yet
module	liquid_biopsy	376	not yet
module	living_computer	857	not yet
module	living_tx	184	not yet
module	metabodesigner	170	not yet
module	micro_tx	218	not yet
module	microaiverse	138	not yet
module	microbiome_exp	119	not yet
module	microbiome_rx	172	not yet

kind	name	py lines	status
module	molecule_eval	149	not yet
module	mutdock	337	not yet
module	neodti_engine	257	not yet
module	neuro_hub	44	not yet
module	neuro_hub_dashboard	25	not yet
module	neuro_pipeline	326	not yet
module	neuroplan_ai	449	not yet
module	nexus_support	299	not yet
module	omega_stats	161	not yet
module	oncocircuit	124	not yet
module	openclinvar	481	not yet
module	organoid_ai	216	not yet
module	organoid_screen	275	not yet
module	pdx_insight	138	not yet
module	phage_designer	75	not yet
module	phage_tx	161	not yet
module	phageforge	87	not yet
module	prime_design	248	not yet
module	promoter_lib	143	not yet
module	proteinPainter	161	not yet
module	qsar_bench	353	not yet
module	rarenet_ai	365	not yet
module	report_studio	213	not yet
module	riboswitch	142	not yet
module	rna_decoder	120	not yet
module	robotic_flow	128	not yet
module	stability_ai	199	not yet
module	str_scope	278	not yet
module	structural_biophysics	512	not yet
module	syn_bio_studio	43	not yet
module	syn_stab_ai	69	not yet
module	synbio_studio	113	not yet
module	synbio_wizard	224	not yet
module	syndroid	231	not yet
module	synlife_evo	278	not yet
module	synthetic_life	197	not yet
module	tissue_eng	193	not yet
module	vector_opt	95	not yet
module	virtual_cell	504	not yet

A Tools Table

Tool	Version	Status/used in
pyhmmer	0.12.3	installed
ViennaRNA	2.7.2	installed
RDKit	2026.3.6	installed
SpliceAI	1.3.1	installed
TensorFlow	2.15.1	installed
Keras	2.15.0	installed
scikit-learn	1.7.2	installed
SciPy	1.15.3	installed
NumPy	1.26.4	installed
Biopython	1.88	installed
twobitreader	4.0.2	installed
AutoDock Vina (python)	1.2.7	installed
Open Babel	3.1.1.23	installed
pytest	9.1.1	installed
pandas	2.3.3	installed
MHCflurry	2.2.1	sweep/neohunter_mhcflurry_cmp.py
CRISPOR reference scorers (vendored)	web ser- vice/vendored	sweep/_crispor_*_ref.py
EBI Clustal Omega REST	web ser- vice/vendored	data/README.md
CPIC API	web ser- vice/vendored	sweep/pgx_cpic_bench.py
NCBI E-utilities	web ser- vice/vendored	src/sc_validate
LOVD3 API	web ser- vice/vendored	discovery/splice_region_vus/lovd_top20.js
MaxEntScan (maxentpy)	0.0.4	discovery/splice_region_vus/maxent_cmp.py
codon-bias	0.5.0	sweep/codon_tai_enc.py
statsmodels	0.15.0	benchmarks/sweep_codon_cai_multispecies (BH-FDR)
PyTorch	2.14.0+cpu	discovery/splice_region_vus/cnn_cv.py, cnn (splice CNN)
Matplotlib	3.10.9	paper/make_figs.py (all figures)
FreeSASA	2.x (python)	sweep/sasa_vs_freesasa.py -i marks/sweep_sasa_freesasa.json
primer3-py	2.3.1	sweep/primer_vs_primer3.py -i marks/sweep_primer_primer3.json; bio.primer.hairpin_dg
DendroPy	5.1.0	sweep/phylo_vs_dendropy.py -i marks/sweep_phylo_dendropy.json
orfipy	0.0.4	sweep/orf_vs_orfipy.py -i marks/sweep_orf_orfipy.json
pysam (htslib)	0.24.1	sweep/vcf_vs_pysam.py -i marks/sweep_vcf_pysam.json (remote t of ClinVar VCF)
PyDESeq2	0.5.2	sweep/de_vs_pydeseq2.py -i marks/sweep_de_pydeseq2.json; bio.de.de_analysis_deseq2
parasail	2.6.1	sweep/align_vs_parasail.py -i marks/sweep_align_parasail.json
g:Profiler g:GOST API	e114.eg62.p19.07.16.04.83	sweep/dex_enrichment_gprofiler.py -i marks/sweep_dex_enrichment.json
STRING API	12.0	sweep/dex_string_ppi.py -i marks/sweep_dex_string_ppi.json
Reactome AnalysisService	live 2026- 09-24	sweep/dex_reactome.py -i marks/sweep_dex_reactome.json
networkx	local 3.4.2	sweep/dex_string_network.py -i marks/sweep_dex_string_network.json
gseapy	local 1.3.1	sweep/dex_gsea_hallmark.py -i

B Dataset Manifest

#	Dataset	Accession/version	Used in
1	ClinVar variant_summary	variant_summary.txt.gz (GRCh38)	data/clinvar_sample_500.tsv; discovery/splice_region_vus/
2	UCSC hg38 reference genome	hg38.2bit	discovery/splice_region_vus/extract.py
3	LOVD3 shared variant records	top-20 VUS query	discovery/splice_region_vus/lovd_top20.json
4	SKEMPI 2.0	skempi_v2.csv (PMID 30598530)	data/skempi_v2.csv
5	BiGG e.coli_core	e.coli_core (PMID 21988831)	data/e.coli_core.json
6	NCBI RefSeqGene BRCA1	NG_005905.2:60000-70000	data/brca1_10kb.fa; data/crispr_fresh_pairs.tsv
7	CRISPOR reference CFD/MIT matrices	crisporWebsite (PMID 30341956)	data/crispr_cfd_oracle.tsv; sweep/_crispor_mit_ref.py
8	CRISPOR Rule Set 1 reference	crisporEffScores.py	data/doench_rs1_oracle.tsv
9	UniProtKB reviewed globins	25 sequences	data/globins25.fa
10	UniProtKB reviewed non-globins	25 sequences	data/nonglobins25.fa
11	Pfam Globin HMM	PF00042.29	data/pfam_PF00042_globin.hmm
12	Rfam tRNA seed	RF00005	data/rfam/RF00005.sto
13	Rfam 5S rRNA seed	RF00001	data/rfam/RF00001.sto
14	Rfam RNase P RNA seed	RF00010	data/rfam/RF00010.sto
15	ChEMBL approved drugs	1000-compound slice	data/chembl_approved_1000.tsv
16	Kazusa CUTG E. coli K12	taxid 83333	data/README.md (sanity only)
17	NCBI RefSeq E. coli K-12 MG1655 CDS	GCF_000005845.2	sweep/codon_cai_bench.py
18	PaxDb E. coli integrated abundance	511145-WHOLE_ORGANISM-integrated	sweep/codon_cai_bench.py
19	CPIC CYP2C19 diplotype table	666 diplotypes	sweep/pgx_cpic_bench.py
20	CPIC CYP2D6 diplotype table	16836 diplotypes	sweep/pgx_cpic_bench.py
21	CPIC CYP2D6 allele activity table	208 alleles	sugarcode-ai pgx_guidelines/data
22	Edinburgh Genome Foundry codon tables	e.coli_316407	sweep/codon_cai_bench.py (via sugarcode-ai)
23	IEDB MHC-I binding data 2013 (Kim et al. 2014)	bdata.20130222.mhci.txt	sweep/neohunter_iedb_bench.py
24	LP-PDBBind (Li et al. leak-proof split)	LP.PDBBind.csv	sweep/docking_bench.py
25	NCBI RefSeq transcripts (TP53, BRCA1, BRCA2, BRAF, EGFR)	NM_000546.6, NM_007294.4, NM_000059.4, NM_004333.6, NM_005228.5	sweep/primer_vs_primer3.py
26	ClinVar GRCh38 VCF (tabix regions TP53, BRCA1, BRCA2, BRAF, EGFR)	clinvar.vcf.gz (46,603 records)	sweep/vcf_vs_pysam.py
27	GEO GSE52778 airway smooth muscle RNA-seq (Himes et al. 2014), NCBI raw counts	GSE52778_raw_counts_GRCh38.p13NCBI raw counts	sweep/ncbi_raw_counts.py

C Accession-Level Dataset Manifest

Each row is one identifier-backed record fetched individually and used in a committed analysis (counting rule in manifests/README.md).

#	Kind	Study	Accession	Used in
1	study	1	variant_summary.txt.gz (GRCh38)	data/clinvar_sample_500.tsv; discovery/splice_region_vus/
2	study	2	hg38.2bit	discovery/splice_region_vus/extract.py
3	study	3	top-20 VUS query	discovery/splice_region_vus/lovd_top20.json
4	study	4	skempi_v2.csv (PMID 30598530)	data/skempi_v2.csv
5	study	5	e.coli_core (PMID 21988831)	data/e.coli_core.json
6	study	6	NG_005905.2:60000-70000	data/brca1_10kb.fa; data/crispr_fresh_pairs.tsv
7	study	7	crisporWebsite (PMID 30341956)	data/crispr_cfd_oracle.tsv; sweep/_crispor_mit_ref.py

#	Kind	Study	Accession	Used in
8	study	8	crisporEffScores.py	data/doench_rsl_oracle.tsv
9	study	9	25 sequences	data/globins25.fa
10	study	10	25 sequences	data/nonglobins25.fa
11	study	11	PF00042.29	data/pfam.PF00042_globin.hmm
12	study	12	RF00005	data/rfam/RF00005.sto
13	study	13	RF00001	data/rfam/RF00001.sto
14	study	14	RF00010	data/rfam/RF00010.sto
15	study	15	1000-compound slice	data/chembl-approved_1000.tsv
16	study	16	taxid 83333	data/README.md (sanity only)
17	study	17	GCF_000005845.2	sweep/codon_cai_bench.py
18	study	18	511145- WHOLE_ORGANISM- integrated	sweep/codon_cai_bench.py
19	study	19	666 diplotypes	sweep/pgx_cplic_bench.py
20	study	20	16836 diplotypes	sweep/pgx_cplic_bench.py
21	study	21	208 alleles	sugarcodes-ai pgx_guidelines/data
22	study	22	e.coli_316407	sweep/codon_cai_bench.py (via sugarcodes-ai)
23	study	23	bdata.20130222.mhci.txt	sweep/neohunter_iedb_bench.py
24	study	24	LP_PDBBind.csv	sweep/docking_bench.py
25	pdb_entry		1CNW	sweep/docking_bench.py
26	pdb_entry		1EJN	sweep/docking_bench.py
27	pdb_entry		1F5L	sweep/docking_bench.py
28	pdb_entry		1G54	sweep/docking_bench.py
29	pdb_entry		1GI1	sweep/docking_bench.py
30	pdb_entry		1K1J	sweep/docking_bench.py
31	pdb_entry		1O3L	sweep/docking_bench.py
32	pdb_entry		1P1O	sweep/docking_bench.py
33	pdb_entry		1Y3X	sweep/docking_bench.py
34	pdb_entry		1ZGI	sweep/docking_bench.py
35	pdb_entry		2ERZ	sweep/docking_bench.py
36	pdb_entry		2J4I	sweep/docking_bench.py
37	pdb_entry		2P4J	sweep/docking_bench.py
38	pdb_entry		2X7T	sweep/docking_bench.py
39	pdb_entry		2XAB	sweep/docking_bench.py
40	pdb_entry		2XJG	sweep/docking_bench.py
41	pdb_entry		2ZDN	sweep/docking_bench.py
42	pdb_entry		3B3C	sweep/docking_bench.py
43	pdb_entry		3C84	sweep/docking_bench.py
44	pdb_entry		3D8W	sweep/docking_bench.py
45	pdb_entry		3HKQ	sweep/docking_bench.py
46	pdb_entry		3HLL	sweep/docking_bench.py
47	pdb_entry		3IBI	sweep/docking_bench.py
48	pdb_entry		3M35	sweep/docking_bench.py
49	pdb_entry		3PO6	sweep/docking_bench.py
50	pdb_entry		3R17	sweep/docking_bench.py
51	pdb_entry		3R4N	sweep/docking_bench.py
52	pdb_entry		4B2I	sweep/docking_bench.py
53	pdb_entry		4DB7	sweep/docking_bench.py
54	pdb_entry		4F5Y	sweep/docking_bench.py
55	pdb_entry		4FL1	sweep/docking_bench.py
56	pdb_entry		4J93	sweep/docking_bench.py
57	pdb_entry		4JFM	sweep/docking_bench.py
58	pdb_entry		4KNJ	sweep/docking_bench.py
59	pdb_entry		4LXZ	sweep/docking_bench.py
60	pdb_entry		4MO8	sweep/docking_bench.py
61	pdb_entry		4NH7	sweep/docking_bench.py
62	pdb_entry		4O04	sweep/docking_bench.py
63	pdb_entry		4OAG	sweep/docking_bench.py
64	pdb_entry		4R76	sweep/docking_bench.py
65	pdb_entry		4RFC	sweep/docking_bench.py
66	pdb_entry		4TIM	sweep/docking_bench.py
67	pdb_entry		4XIT	sweep/docking_bench.py
68	pdb_entry		4YTC	sweep/docking_bench.py
69	pdb_entry		4YXI	sweep/docking_bench.py
70	pdb_entry		4ZME	sweep/docking_bench.py
71	pdb_entry		5AML	sweep/docking_bench.py

#	Kind	Study	Accession	Used in
72	pdb_entry		5CAP	sweep/docking_bench.py
73	pdb_entry		5CAU	sweep/docking_bench.py
74	pdb_entry		5D3L	sweep/docking_bench.py
75	pdb_entry		5EWK	sweep/docking_bench.py
76	pdb_entry		5EWY	sweep/docking_bench.py
77	pdb_entry		5F60	sweep/docking_bench.py
78	pdb_entry		5F62	sweep/docking_bench.py
79	pdb_entry		5G5V	sweep/docking_bench.py
80	pdb_entry		5HTZ	sweep/docking_bench.py
81	pdb_entry		5IX0	sweep/docking_bench.py
82	pdb_entry		5L8C	sweep/docking_bench.py
83	pdb_entry		5MJN	sweep/docking_bench.py
84	pdb_entry		5MPN	sweep/docking_bench.py
85	pdb_entry		5N24	sweep/docking_bench.py
86	pdb_entry		5O87	sweep/docking_bench.py
87	pdb_entry		5SZ4	sweep/docking_bench.py
88	pdb_entry		5SZ5	sweep/docking_bench.py
89	pdb_entry		5Y8Y	sweep/docking_bench.py
90	pdb_entry		6EQ5	sweep/docking_bench.py
91	pdb_entry		6G38	sweep/docking_bench.py
92	pdb_entry		6G6T	sweep/docking_bench.py
93	pdb_entry		6HQY	sweep/docking_bench.py
94	pdb_entry		6HRQ	sweep/docking_bench.py
95	pdb_entry		6OE0	sweep/docking_bench.py
96	pdb_entry		6RFN	sweep/docking_bench.py
97	pdb_entry		6U6W	sweep/docking_bench.py
98	species		10090-ASTROCYTE-integrated.txt	sweep/codon_cai_multispecies.py
99	species		GCF_000001635.27	sweep/codon_cai_multispecies.py
100	species		10116-CARDIAC_VENTRICLE-integrated.txt	sweep/codon_cai_multispecies.py
101	species		GCF_036323735.1	sweep/codon_cai_multispecies.py
102	species		121845-WHOLE_ORGANISM-integrated.txt	sweep/codon_cai_multispecies.py
103	species		GCF_000475195.1	sweep/codon_cai_multispecies.py
104	species		1295-WHOLE_ORGANISM-integrated.txt	sweep/codon_cai_multispecies.py
105	species		GCF_025558325.1	sweep/codon_cai_multispecies.py
106	species		1314-WHOLE_ORGANISM-integrated.txt	sweep/codon_cai_multispecies.py
107	species		GCF_900475035.1	sweep/codon_cai_multispecies.py
108	species		185431-WHOLE_ORGANISM-integrated.txt	sweep/codon_cai_multispecies.py
109	species		GCF_000002445.2	sweep/codon_cai_multispecies.py
110	species		192222-WHOLE_ORGANISM-integrated.txt	sweep/codon_cai_multispecies.py
111	species		GCF_000009085.1	sweep/codon_cai_multispecies.py
112	species		196627-WHOLE_ORGANISM-integrated.txt	sweep/codon_cai_multispecies.py
113	species		GCF_000011325.1	sweep/codon_cai_multispecies.py
114	species		198214-WHOLE_ORGANISM-integrated.txt	sweep/codon_cai_multispecies.py
115	species		GCF_000006925.2	sweep/codon_cai_multispecies.py
116	species		208964-WHOLE_ORGANISM-integrated.txt	sweep/codon_cai_multispecies.py
117	species		GCF_000006765.1	sweep/codon_cai_multispecies.py

#	Kind	Study	Accession	Used in
118	species		224308- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
119	species		GCF_000009045.1	sweep/codon_cai_multispecies.py
120	species		224911- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
121	species		GCF_0001642675.1	sweep/codon_cai_multispecies.py
122	species		243159- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
123	species		GCF_000021485.1	sweep/codon_cai_multispecies.py
124	species		243230- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
125	species		GCF_020546685.1	sweep/codon_cai_multispecies.py
126	species		246200- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
127	species		GCF_000011965.2	sweep/codon_cai_multispecies.py
128	species		2903- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
129	species		GCF_000372725.1	sweep/codon_cai_multispecies.py
130	species		343509- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
131	species		GCF_000010085.1	sweep/codon_cai_multispecies.py
132	species		36329- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
133	species		GCF_000002765.6	sweep/codon_cai_multispecies.py
134	species		403677- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
135	species		GCF_000142945.1	sweep/codon_cai_multispecies.py
136	species		4081-PERICARP- integrated.txt	sweep/codon_cai_multispecies.py
137	species		GCF_036512215.1	sweep/codon_cai_multispecies.py
138	species		4097-LEAF-integrated.txt	sweep/codon_cai_multispecies.py
139	species		GCF_000715075.1	sweep/codon_cai_multispecies.py
140	species		418985- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
141	species		GCA_002081605.1	sweep/codon_cai_multispecies.py
142	species		44689- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
143	species		GCF_000004695.1	sweep/codon_cai_multispecies.py
144	species		4932- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
145	species		GCF_000146045.2	sweep/codon_cai_multispecies.py
146	species		54291- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
147	species		GCF_030505655.1	sweep/codon_cai_multispecies.py
148	species		6239- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
149	species		GCF_000002985.6	sweep/codon_cai_multispecies.py
150	species		7227-SPERM-integrated.txt	sweep/codon_cai_multispecies.py
151	species		GCF_000001215.4	sweep/codon_cai_multispecies.py
152	species		7460-ABDOMEN- integrated.txt	sweep/codon_cai_multispecies.py
153	species		GCF_003254395.2	sweep/codon_cai_multispecies.py
154	species		8030-LIVER-integrated.txt	sweep/codon_cai_multispecies.py

#	Kind	Study	Accession	Used in
155	species		GCF.965601325.2	sweep/codon_cai_multispecies.py
156	species		83332- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
157	species		GCF.000195955.2	sweep/codon_cai_multispecies.py
158	species		9031-HEART-integrated.txt	sweep/codon_cai_multispecies.py
159	species		GCF.016699485.2	sweep/codon_cai_multispecies.py
160	species		93061- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
161	species		GCF.000013425.1	sweep/codon_cai_multispecies.py
162	species		9544-BILE_DUCT- integrated.txt	sweep/codon_cai_multispecies.py
163	species		GCF.049350105.2	sweep/codon_cai_multispecies.py
164	species		9606-ADRENAL_GLAND- integrated.txt	sweep/codon_cai_multispecies.py
165	species		GCF.000001405.40	sweep/codon_cai_multispecies.py
166	species		9615-PLASMA- integrated.txt	sweep/codon_cai_multispecies.py
167	species		GCF.011100685.1	sweep/codon_cai_multispecies.py
168	species		9823-LIVER-integrated.txt	sweep/codon_cai_multispecies.py
169	species		GCF.054392235.1	sweep/codon_cai_multispecies.py
170	species		9913- LENS_OF_CAMERA_TYPE_EYE- integrated.txt	sweep/codon_cai_multispecies.py
171	species		GCF.002263795.3	sweep/codon_cai_multispecies.py
172	species		99287- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
173	species		GCF.000006945.2	sweep/codon_cai_multispecies.py
174	species		9986- PERIPHERAL_REGION_OF_RETINA- integrated.txt	sweep/codon_cai_multispecies.py
175	species		GCF.964237555.1	sweep/codon_cai_multispecies.py
176	gtrnadb		Esch.coli_K_12_MG1655	sweep/codon_tai_enc.py
177	gtrnadb		Baci.subt_subtilis_168	sweep/codon_tai_enc.py
178	gtrnadb		Scere3	sweep/codon_tai_enc.py
179	refseq_transcript	25	NM.000546.6	sweep/primer_vs_primer3.py
180	refseq_transcript	25	NM.007294.4	sweep/primer_vs_primer3.py
181	refseq_transcript	25	NM.000059.4	sweep/primer_vs_primer3.py
182	refseq_transcript	25	NM.004333.6	sweep/primer_vs_primer3.py
183	refseq_transcript	25	NM.005228.5	sweep/primer_vs_primer3.py
184	study	26	clinvar.vcf.gz	sweep/vcf_vs_pysam.py
185	study	27	GSE52778	sweep/de_vs_pydeseq2.py
186	rfam_family	RF00004	RF00004	sweep/stockholm_vs_biopython.py
187	rfam_family	RF00015	RF00015	sweep/stockholm_vs_biopython.py
188	rfam_family	RF00017	RF00017	sweep/stockholm_vs_biopython.py
189	rfam_family	RF00023	RF00023	sweep/stockholm_vs_biopython.py
190	rfam_family	RF00050	RF00050	sweep/stockholm_vs_biopython.py
191	rfam_family	RF00059	RF00059	sweep/stockholm_vs_biopython.py
192	rfam_family	RF00162	RF00162	sweep/stockholm_vs_biopython.py
193	rfam_family	RF00167	RF00167	sweep/stockholm_vs_biopython.py
194	rfam_family	RF00168	RF00168	sweep/stockholm_vs_biopython.py
195	rfam_family	RF00174	RF00174	sweep/stockholm_vs_biopython.py
196	rfam_family	RF00234	RF00234	sweep/stockholm_vs_biopython.py
197	rfam_family	RF00504	RF00504	sweep/stockholm_vs_biopython.py
198	refseq_genome	J01749.1	J01749.1	sweep/proteinprops_restriction_vs_biopython.py
199	refseq_genome	NC_001416.1	NC_001416.1	sweep/proteinprops_restriction_vs_biopython.py
200	opentree_node	ott1062253	ott1062253	sweep/newick_vs_biophylo.py
201	opentree_node	ott392223	ott392223	sweep/newick_vs_biophylo.py
202	opentree_node	ott44565	ott44565	sweep/newick_vs_biophylo.py
203	opentree_node	ott691846	ott691846	sweep/newick_vs_biophylo.py
204	opentree_node	ott81461	ott81461	sweep/newick_vs_biophylo.py
205	opentree_node	ott913935	ott913935	sweep/newick_vs_biophylo.py
206	study	NA12878_chr20_lowcov	NA12878.chrom20.ILLUMINA	sweep/fit_low_pysam.py
207	study	SRR622461	SRR622461	sweep/fastq_vs_biopython.py
208	jaspar_matrix	MA0003.4	MA0003.4	sweep/motif_vs_biopython.py

#	Kind	Study	Accession	Used in
209	jaspar_matrix	MA0079.3	MA0079.3	sweep/motif_vs.biopython.py
210	jaspar_matrix	MA0080.4	MA0080.4	sweep/motif_vs.biopython.py
211	jaspar_matrix	MA0099.3	MA0099.3	sweep/motif_vs.biopython.py
212	jaspar_matrix	MA0105.4	MA0105.4	sweep/motif_vs.biopython.py
213	jaspar_matrix	MA0106.3	MA0106.3	sweep/motif_vs.biopython.py
214	jaspar_matrix	MA0108.2	MA0108.2	sweep/motif_vs.biopython.py
215	jaspar_matrix	MA0112.3	MA0112.3	sweep/motif_vs.biopython.py
216	jaspar_matrix	MA0113.3	MA0113.3	sweep/motif_vs.biopython.py
217	jaspar_matrix	MA0139.1	MA0139.1	sweep/motif_vs.biopython.py
218	nucore_record	NC_000913.3	NC_000913.3	sweep/genbank_vs.biopython.py
219	nucore_record	NC_001422.1	NC_001422.1	sweep/genbank_vs.biopython.py
220	nucore_record	NC_001477.1	NC_001477.1	sweep/genbank_vs.biopython.py
221	nucore_record	NC_001498.1	NC_001498.1	sweep/genbank_vs.biopython.py
222	nucore_record	NC_001802.1	NC_001802.1	sweep/genbank_vs.biopython.py
223	nucore_record	NC_001806.2	NC_001806.2	sweep/genbank_vs.biopython.py
224	nucore_record	NC_002549.1	NC_002549.1	sweep/genbank_vs.biopython.py
225	nucore_record	NC_003977.2	NC_003977.2	sweep/genbank_vs.biopython.py
226	nucore_record	NC_012920.1	NC_012920.1	sweep/genbank_vs.biopython.py
227	nucore_record	NC_045512.2	NC_045512.2	sweep/genbank_vs.biopython.py
228	nucore_record	NG_005905.2	NG_005905.2	sweep/genbank_vs.biopython.py
229	nucore_record	NM_000038.6	NM_000038.6	sweep/genbank_vs.biopython.py
230	nucore_record	NM_000249.4	NM_000249.4	sweep/genbank_vs.biopython.py
231	nucore_record	NM_000251.3	NM_000251.3	sweep/genbank_vs.biopython.py
232	nucore_record	NM_000314.8	NM_000314.8	sweep/genbank_vs.biopython.py
233	nucore_record	NM_001126112.3	NM_001126112.3	sweep/genbank_vs.biopython.py
234	nucore_record	NM_001904.4	NM_001904.4	sweep/genbank_vs.biopython.py
235	nucore_record	NM_002524.5	NM_002524.5	sweep/genbank_vs.biopython.py
236	nucore_record	NM_004985.5	NM_004985.5	sweep/genbank_vs.biopython.py
237	study	29	hg38.cpgIslandExt.chr20	sweep/bed_vs.bioframe.py
238	study	30	hg38.rmsk.chr20:0-500000	sweep/bed_vs.bioframe.py
239	nucore_record	MG772933.1	MG772933.1	sweep/kmer_vs.sourmash.py
240	nucore_record	MN996532.2	MN996532.2	sweep/kmer_vs.sourmash.py
241	nucore_record	MT040333.1	MT040333.1	sweep/kmer_vs.sourmash.py
242	nucore_record	NC_004718.3	NC_004718.3	sweep/kmer_vs.sourmash.py
243	nucore_record	NC_019843.3	NC_019843.3	sweep/kmer_vs.sourmash.py
244	db SNP_variant	rs1042713	rs1042713	sweep/gstats_vs.scipy.py
245	db SNP_variant	rs10427255	rs10427255	sweep/gstats_vs.scipy.py
246	db SNP_variant	rs1229984	rs1229984	sweep/gstats_vs.scipy.py
247	db SNP_variant	rs12913832	rs12913832	sweep/gstats_vs.scipy.py
248	db SNP_variant	rs1426654	rs1426654	sweep/gstats_vs.scipy.py
249	db SNP_variant	rs16891982	rs16891982	sweep/gstats_vs.scipy.py
250	db SNP_variant	rs17822931	rs17822931	sweep/gstats_vs.scipy.py
251	db SNP_variant	rs1799945	rs1799945	sweep/gstats_vs.scipy.py
252	db SNP_variant	rs1800562	rs1800562	sweep/gstats_vs.scipy.py
253	db SNP_variant	rs1801133	rs1801133	sweep/gstats_vs.scipy.py
254	db SNP_variant	rs1815739	rs1815739	sweep/gstats_vs.scipy.py
255	db SNP_variant	rs2231142	rs2231142	sweep/gstats_vs.scipy.py
256	db SNP_variant	rs334	rs334	sweep/gstats_vs.scipy.py
257	db SNP_variant	rs3827760	rs3827760	sweep/gstats_vs.scipy.py
258	db SNP_variant	rs429358	rs429358	sweep/gstats_vs.scipy.py
259	db SNP_variant	rs4680	rs4680	sweep/gstats_vs.scipy.py
260	db SNP_variant	rs4988235	rs4988235	sweep/gstats_vs.scipy.py
261	db SNP_variant	rs53576	rs53576	sweep/gstats_vs.scipy.py
262	db SNP_variant	rs6025	rs6025	sweep/gstats_vs.scipy.py
263	db SNP_variant	rs671	rs671	sweep/gstats_vs.scipy.py
264	db SNP_variant	rs7412	rs7412	sweep/gstats_vs.scipy.py
265	db SNP_variant	rs762551	rs762551	sweep/gstats_vs.scipy.py
266	hgnc_record	HGNC:ABCC11	HGNC:14639	sweep/uniprot_vs.hgnc.py
267	hgnc_record	HGNC:ABL1	HGNC:76	sweep/uniprot_vs.hgnc.py
268	hgnc_record	HGNC:ACTN3	HGNC:165	sweep/uniprot_vs.hgnc.py
269	hgnc_record	HGNC:ADH1B	HGNC:250	sweep/uniprot_vs.hgnc.py
270	hgnc_record	HGNC:ALDH2	HGNC:404	sweep/uniprot_vs.hgnc.py
271	hgnc_record	HGNC:APC	HGNC:583	sweep/uniprot_vs.hgnc.py
272	hgnc_record	HGNC:APOE	HGNC:613	sweep/uniprot_vs.hgnc.py
273	hgnc_record	HGNC:ASXL1	HGNC:18318	sweep/uniprot_vs.hgnc.py
274	hgnc_record	HGNC:ATM	HGNC:795	sweep/uniprot_vs.hgnc.py

#	Kind	Study	Accession	Used in
275	hgnc_record	HGNC:BRAF	HGNC:1097	sweep/uniprot_vs_hgnc.py
276	hgnc_record	HGNC:BRCA1	HGNC:1100	sweep/uniprot_vs_hgnc.py
277	hgnc_record	HGNC:BRCA2	HGNC:1101	sweep/uniprot_vs_hgnc.py
278	hgnc_record	HGNC:CDKN2A	HGNC:1787	sweep/uniprot_vs_hgnc.py
279	hgnc_record	HGNC:CFTR	HGNC:1884	sweep/uniprot_vs_hgnc.py
280	hgnc_record	HGNC:CHEK2	HGNC:16627	sweep/uniprot_vs_hgnc.py
281	hgnc_record	HGNC:COMT	HGNC:2228	sweep/uniprot_vs_hgnc.py
282	hgnc_record	HGNC:DMD	HGNC:2928	sweep/uniprot_vs_hgnc.py
283	hgnc_record	HGNC:DNMT3A	HGNC:2978	sweep/uniprot_vs_hgnc.py
284	hgnc_record	HGNC:EDAR	HGNC:2895	sweep/uniprot_vs_hgnc.py
285	hgnc_record	HGNC:EGFR	HGNC:3236	sweep/uniprot_vs_hgnc.py
286	hgnc_record	HGNC:ERBB2	HGNC:3430	sweep/uniprot_vs_hgnc.py
287	hgnc_record	HGNC:F5	HGNC:3542	sweep/uniprot_vs_hgnc.py
288	hgnc_record	HGNC:FLT3	HGNC:3765	sweep/uniprot_vs_hgnc.py
289	hgnc_record	HGNC:FMR1	HGNC:3775	sweep/uniprot_vs_hgnc.py
290	hgnc_record	HGNC:GBA1	HGNC:4177	sweep/uniprot_vs_hgnc.py
291	hgnc_record	HGNC:HBB	HGNC:4827	sweep/uniprot_vs_hgnc.py
292	hgnc_record	HGNC:HERC2	HGNC:4868	sweep/uniprot_vs_hgnc.py
293	hgnc_record	HGNC:HFE	HGNC:4886	sweep/uniprot_vs_hgnc.py
294	hgnc_record	HGNC:HTT	HGNC:4851	sweep/uniprot_vs_hgnc.py
295	hgnc_record	HGNC:IDH1	HGNC:5382	sweep/uniprot_vs_hgnc.py
296	hgnc_record	HGNC:IDH2	HGNC:5383	sweep/uniprot_vs_hgnc.py
297	hgnc_record	HGNC:JAK2	HGNC:6192	sweep/uniprot_vs_hgnc.py
298	hgnc_record	HGNC:KCNQ1	HGNC:6294	sweep/uniprot_vs_hgnc.py
299	hgnc_record	HGNC:KRAS	HGNC:6407	sweep/uniprot_vs_hgnc.py
300	hgnc_record	HGNC:LCT	HGNC:6530	sweep/uniprot_vs_hgnc.py
301	hgnc_record	HGNC:LMNA	HGNC:6636	sweep/uniprot_vs_hgnc.py
302	hgnc_record	HGNC:MLH1	HGNC:7127	sweep/uniprot_vs_hgnc.py
303	hgnc_record	HGNC:MSH2	HGNC:7325	sweep/uniprot_vs_hgnc.py
304	hgnc_record	HGNC:MTHFR	HGNC:7436	sweep/uniprot_vs_hgnc.py
305	hgnc_record	HGNC:MYC	HGNC:7553	sweep/uniprot_vs_hgnc.py
306	hgnc_record	HGNC:MYH7	HGNC:7577	sweep/uniprot_vs_hgnc.py
307	hgnc_record	HGNC:NF1	HGNC:7765	sweep/uniprot_vs_hgnc.py
308	hgnc_record	HGNC:NF2	HGNC:7773	sweep/uniprot_vs_hgnc.py
309	hgnc_record	HGNC:NPM1	HGNC:7910	sweep/uniprot_vs_hgnc.py
310	hgnc_record	HGNC:NRAS	HGNC:7989	sweep/uniprot_vs_hgnc.py
311	hgnc_record	HGNC:OCA2	HGNC:8101	sweep/uniprot_vs_hgnc.py
312	hgnc_record	HGNC:PALB2	HGNC:26144	sweep/uniprot_vs_hgnc.py
313	hgnc_record	HGNC:PIK3CA	HGNC:8975	sweep/uniprot_vs_hgnc.py
314	hgnc_record	HGNC:PTEN	HGNC:9588	sweep/uniprot_vs_hgnc.py
315	hgnc_record	HGNC:RB1	HGNC:9884	sweep/uniprot_vs_hgnc.py
316	hgnc_record	HGNC:RUNX1	HGNC:10471	sweep/uniprot_vs_hgnc.py
317	hgnc_record	HGNC:SCN5A	HGNC:10593	sweep/uniprot_vs_hgnc.py
318	hgnc_record	HGNC:SLC24A5	HGNC:20611	sweep/uniprot_vs_hgnc.py
319	hgnc_record	HGNC:SMAD4	HGNC:6770	sweep/uniprot_vs_hgnc.py
320	hgnc_record	HGNC:SOD1	HGNC:11179	sweep/uniprot_vs_hgnc.py
321	hgnc_record	HGNC:STK11	HGNC:11389	sweep/uniprot_vs_hgnc.py
322	hgnc_record	HGNC:TET2	HGNC:25941	sweep/uniprot_vs_hgnc.py
323	hgnc_record	HGNC:TP53	HGNC:11998	sweep/uniprot_vs_hgnc.py
324	hgnc_record	HGNC:TTN	HGNC:12403	sweep/uniprot_vs_hgnc.py
325	hgnc_record	HGNC:VHL	HGNC:12687	sweep/uniprot_vs_hgnc.py

References

- [1] Doench JG et al. Optimized sgRNA design to maximize activity and minimize off-target effects of CRISPR-Cas9. *Nat Biotechnol* 34:184-191, 2016.
- [2] Concordet JP, Haeussler M. CRISPOR: intuitive guide selection for CRISPR/Cas9 genome editing experiments and screens. *Nucleic Acids Res* 46:W242-W245, 2018.
- [3] Jankauskaitė J et al. SKEMPI 2.0: an updated benchmark of changes in protein-protein binding energy, kinetics and thermodynamics upon mutation. *Bioinformatics* 35:462-469, 2019.

- [4] King ZA et al. BiGG Models: a platform for integrating, standardizing and sharing genome-scale models. *Nucleic Acids Res* 44:D515-522, 2016.
- [5] Sievers F et al. Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal Omega. *Mol Syst Biol* 7:539, 2011.
- [6] Tavtigian SV et al. Modeling the ACMG/AMP variant classification guidelines as a Bayesian classification framework. *Genet Med* 20:1054-1060, 2018.
- [7] Yeo G, Burge CB. Maximum entropy modeling of short sequence motifs with applications to RNA splicing signals. *J Comput Biol* 11:377-394 (2004).
- [8] Sharp PM, Li WH. The codon adaptation index: a measure of directional synonymous codon usage bias. *Nucleic Acids Res* 15:1281-1295 (1987).
- [9] dos Reis M, Savva R, Wernisch L. Solving the riddle of codon usage preferences: a test for translational selection. *Nucleic Acids Res* 32:5036-5044 (2004).
- [10] Eberhardt J, Santos-Martins D, Tillack AF, Forli S. AutoDock Vina 1.2.0. *J Chem Inf Model* 61:3891-3898 (2021).
- [11] Wang M et al. PaxDb, a database of protein abundance averages across all three domains of life. *Mol Cell Proteomics* 11:492-500 (2012).